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(54) **SYSTEMS, DEVICES, AND METHODS FOR COMBINED CELL PROCESSES**

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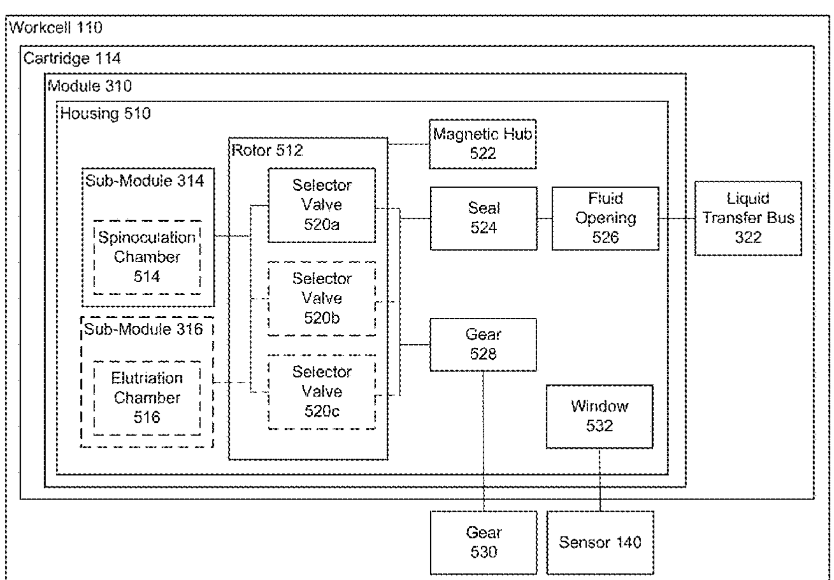
(57) **ABSTRACT**

The present disclosure relates to systems, devices, and methods for spinoculation and counterflow centrifugal elutriation. In an embodiment, the present disclosure relates to a cartridge in an automated system, comprising a liquid transfer bus, a module fluidically coupled to the liquid transfer bus, the module comprising two sub-modules, each configured to perform separate cell processing steps, and at least one selector valve configured to direct fluid to at least one of the sub-modules.

24 Claims, 15 Drawing Sheets



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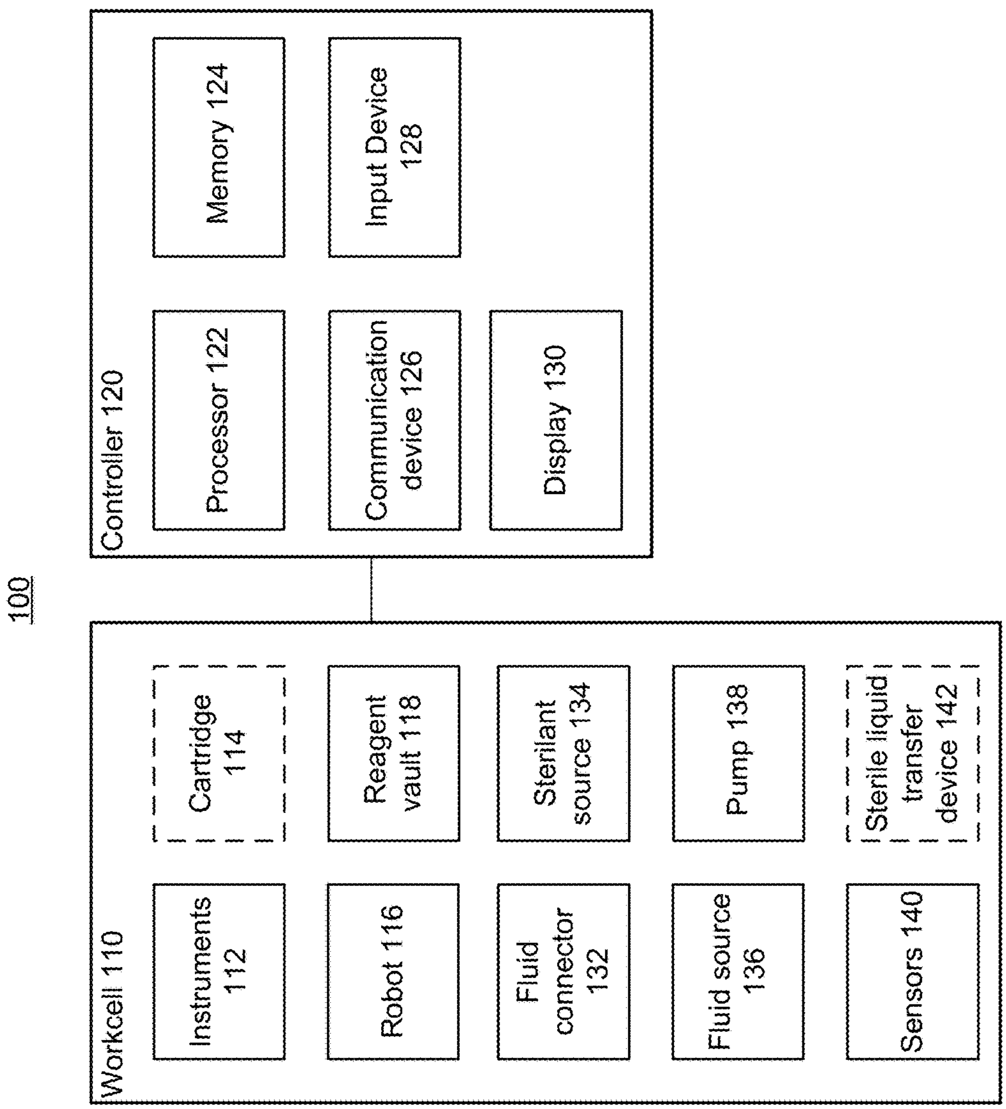

100
Controller 120
Processor 122
Memory 124
Communication device 126
Input Device 128
Display 130
Workcell 110
Instruments 112
Cartridge 114
Robot 116
Reagent vault 118
Fluid connector 132
Sterilant source 134
Fluid source 136
Pump 138
Sensors 140
Sterile liquid transfer device 142

FIG. 1

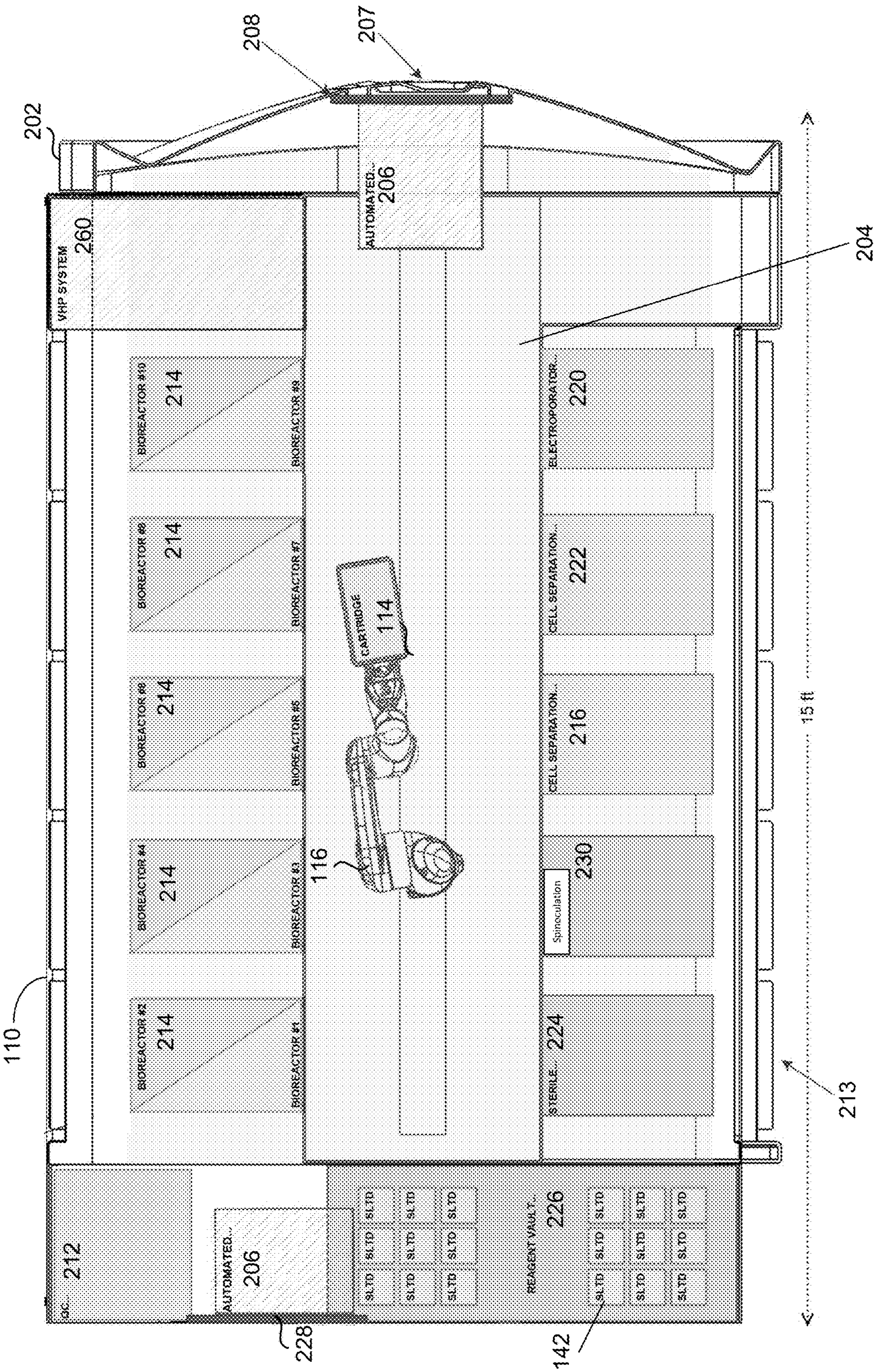

202
208
207
AUTOMATED...
206
VHP SYSTEM
260
204
BIOREACTOR #10
214
BIOREACTOR #9
BIOREACTOR #8
214
BIOREACTOR #7
BIOREACTOR #6
214
BIOREACTOR #5
BIOREACTOR #4
214
BIOREACTOR #3
BIOREACTOR #2
214
BIOREACTOR #1
ELECTROPORATOR...
220
CARTRIDGE
114
CELL SEPARATION...
222
CELL SEPARATION...
216
116
Spinoculation
230
STERILE...
224
110
213
15 ft
QC
212
AUTOMATED...
206
REAGENT VAULT...
226
SLTD
228
142

FIG. 2A

FIG. 2B

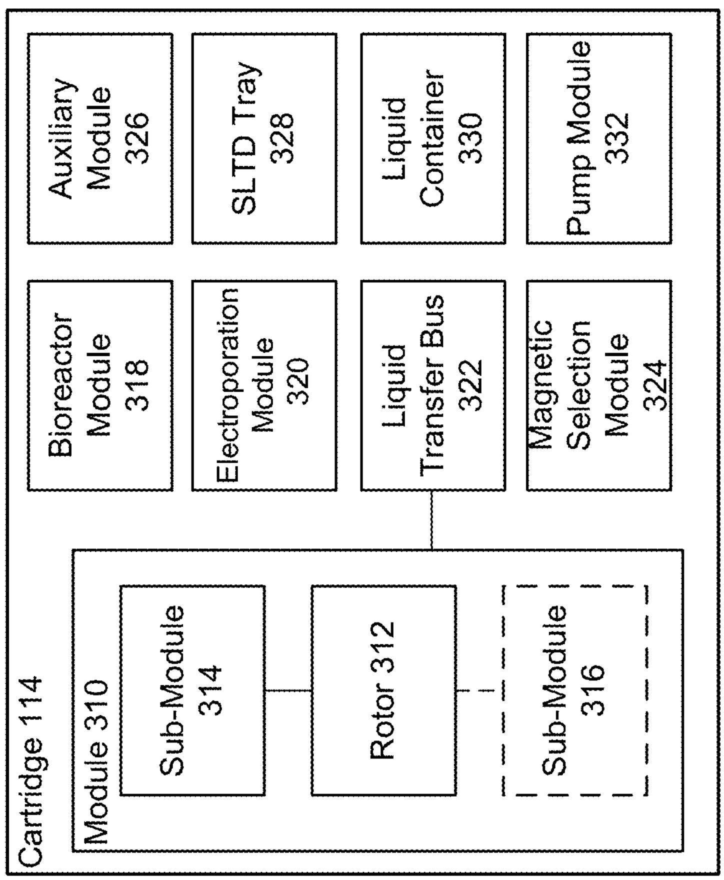

Cartridge 114
Module 310
Sub-Module 314
Rotor 312
Sub-Module 316
Bioreactor Module 318
Electroporation Module 320
Liquid Transfer Bus 322
Magnetic Selection Module 324
Auxiliary Module 326
SLTD Tray 328
Liquid Container 330
Pump Module 332

FIG. 3

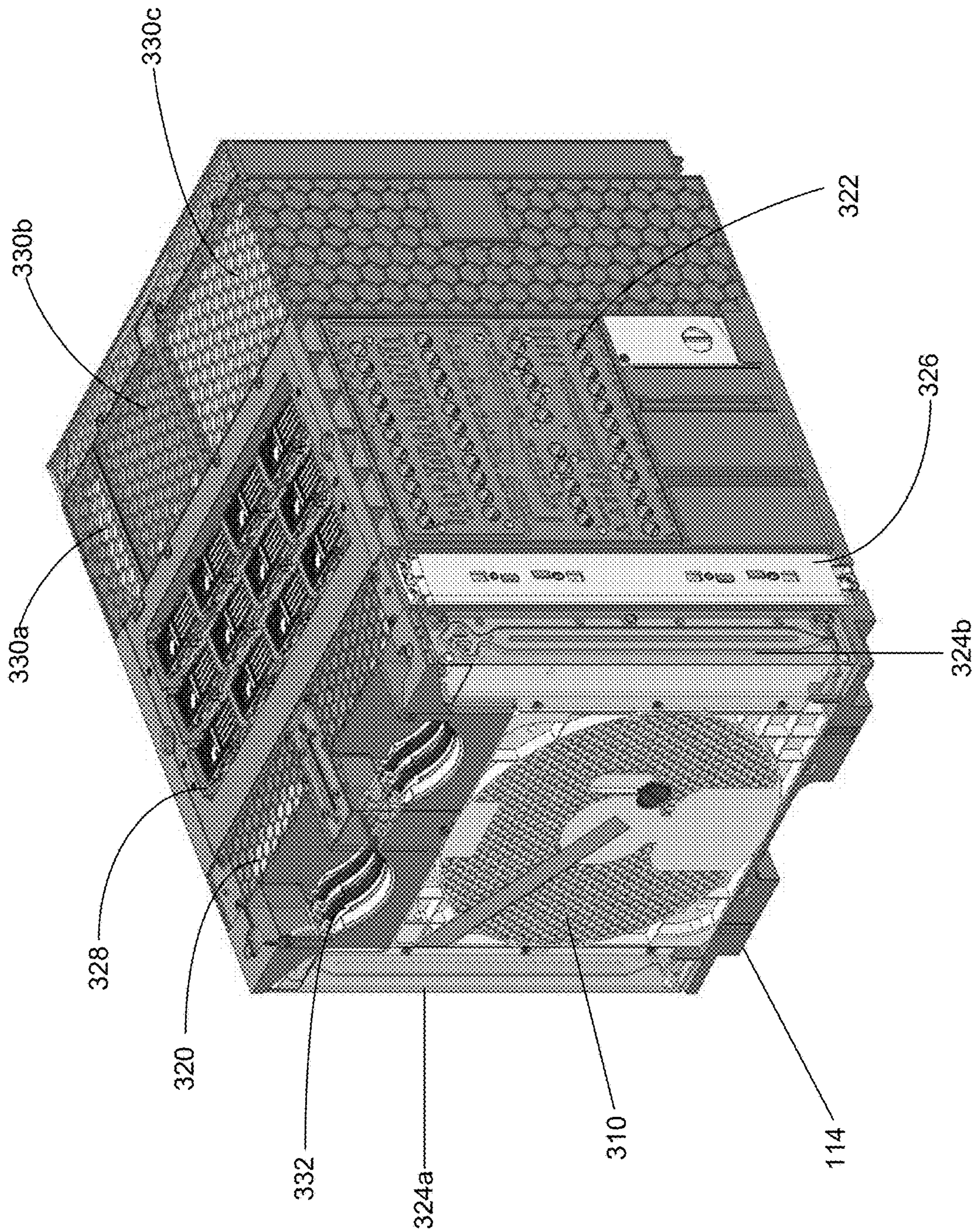


FIG. 4

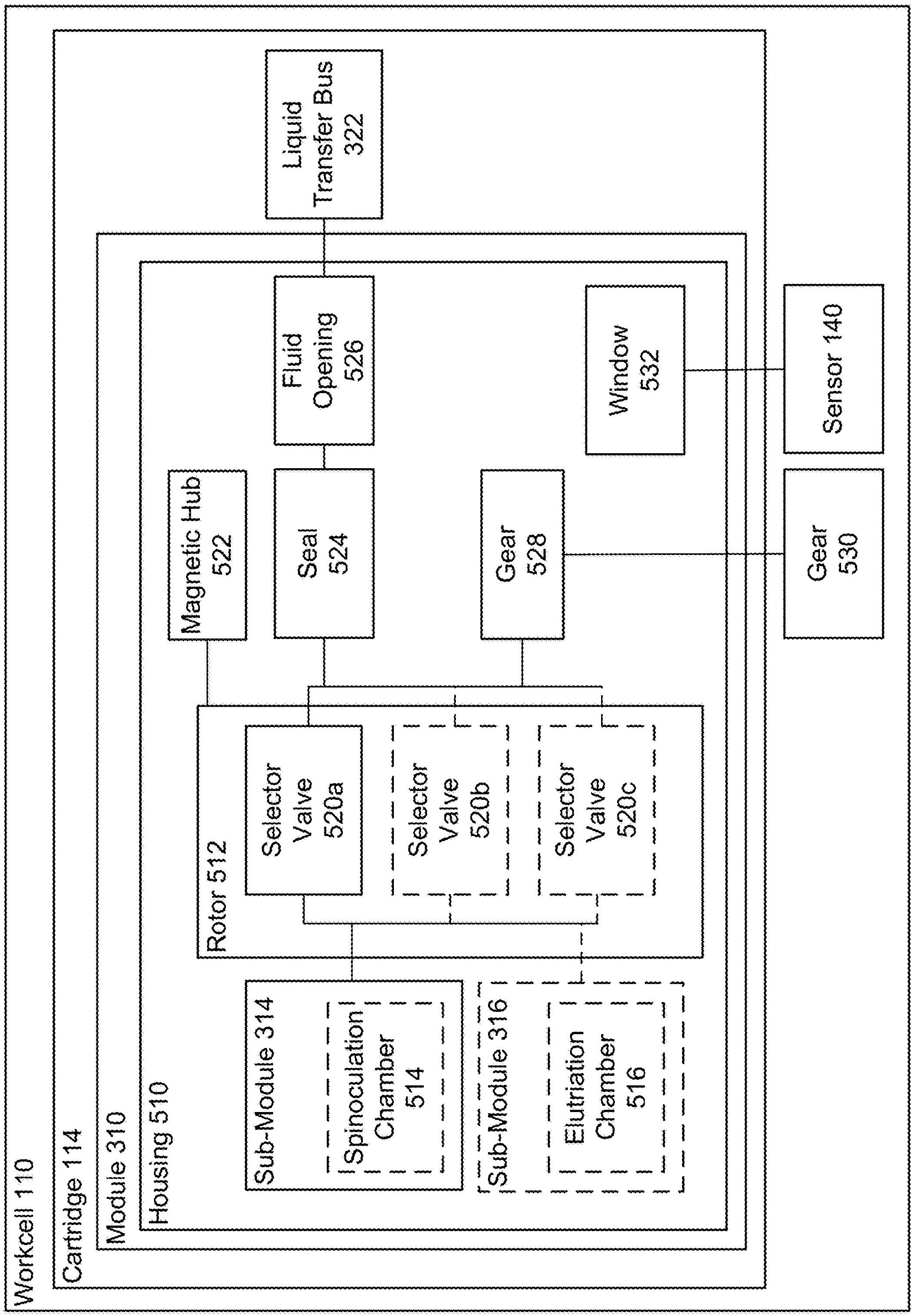

Workcell 110
Cartridge 114
Module 310
Housing 510
Liquid Transfer Bus 322
Fluid Opening 526
Window 532
Sensor 140
Magnetic Hub 522
Seal 524
Gear 528
Gear 530
Rotor 512
Selector Valve 520a
Selector Valve 520b
Selector Valve 520c
Sub-Module 314
Spinoculation Chamber 514
Sub-Module 316
Elutriation Chamber 516

FIG. 5

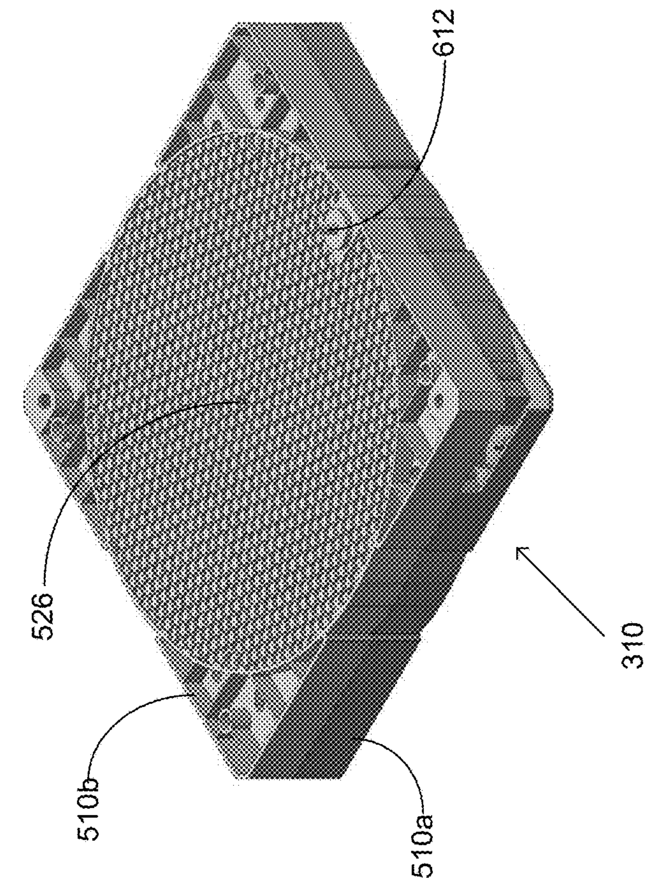


FIG. 6B

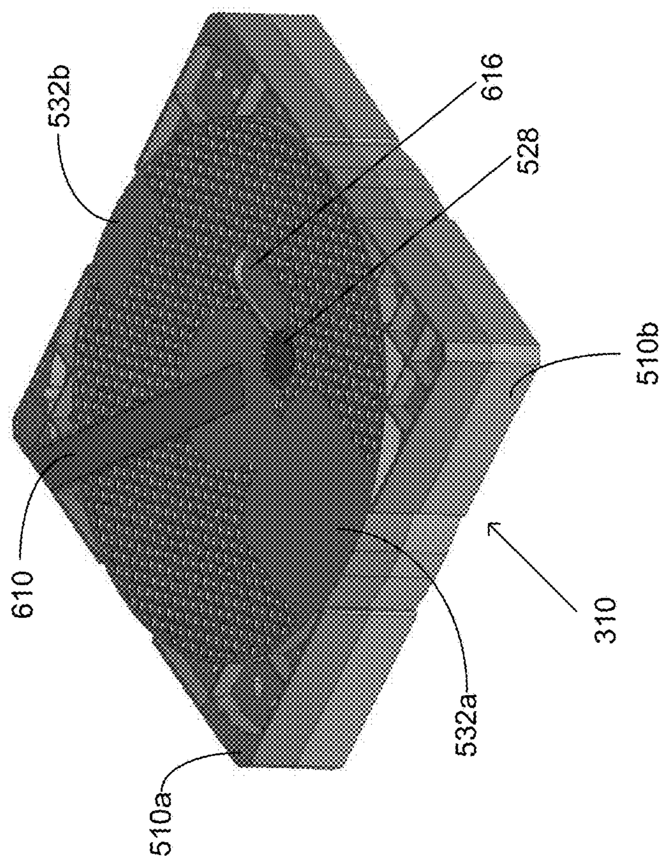


FIG. 6A

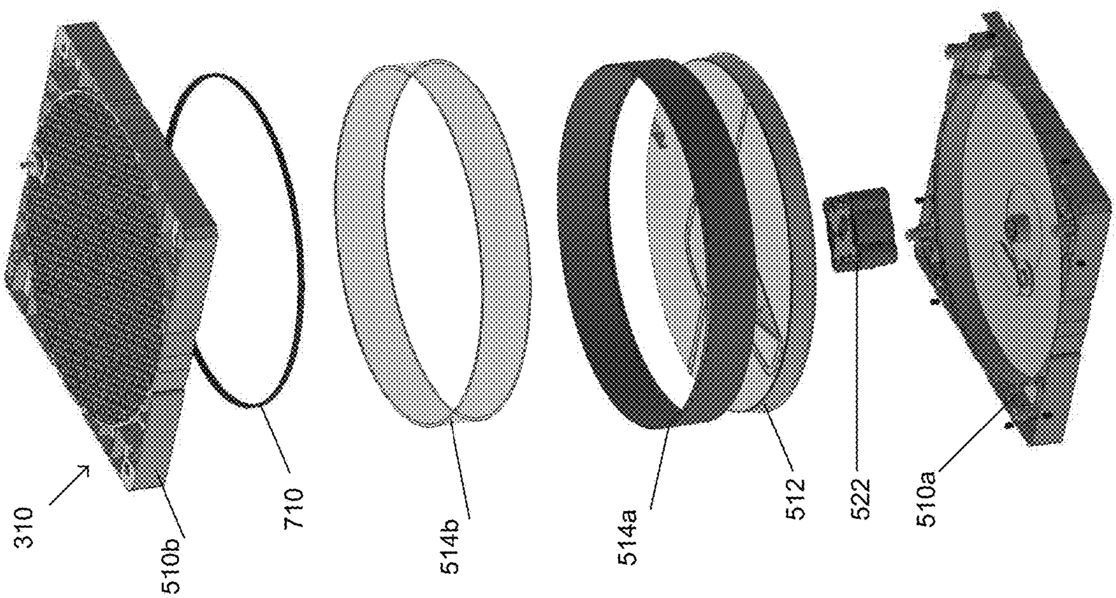

310
510b
710
514b
514a
512
522
510a

FIG. 7B

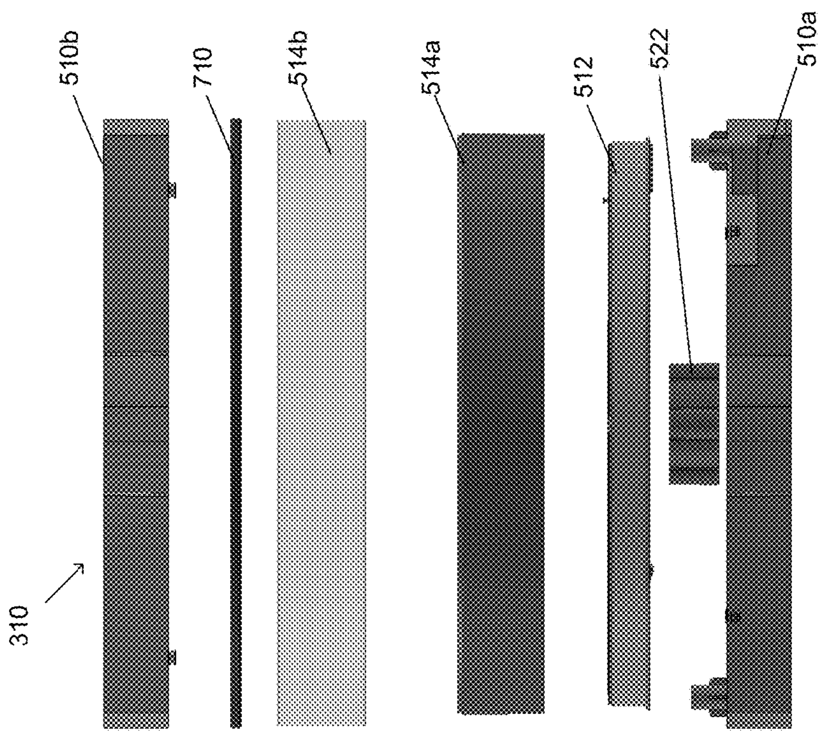

310
510b
710
514b
514a
512
522
510a

FIG. 7A

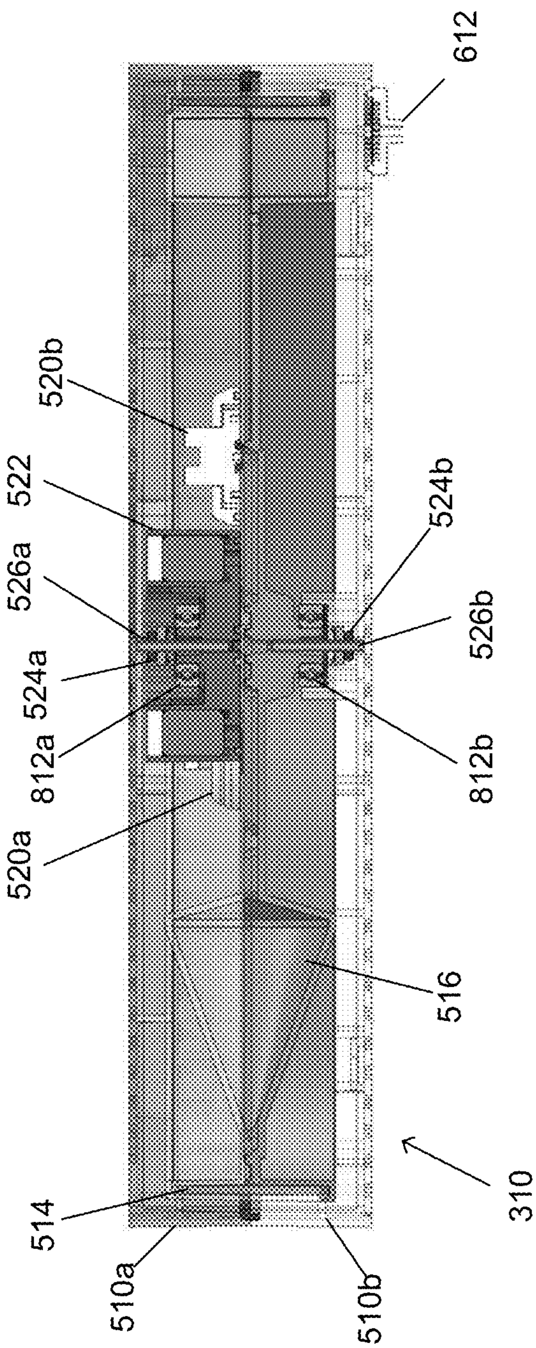
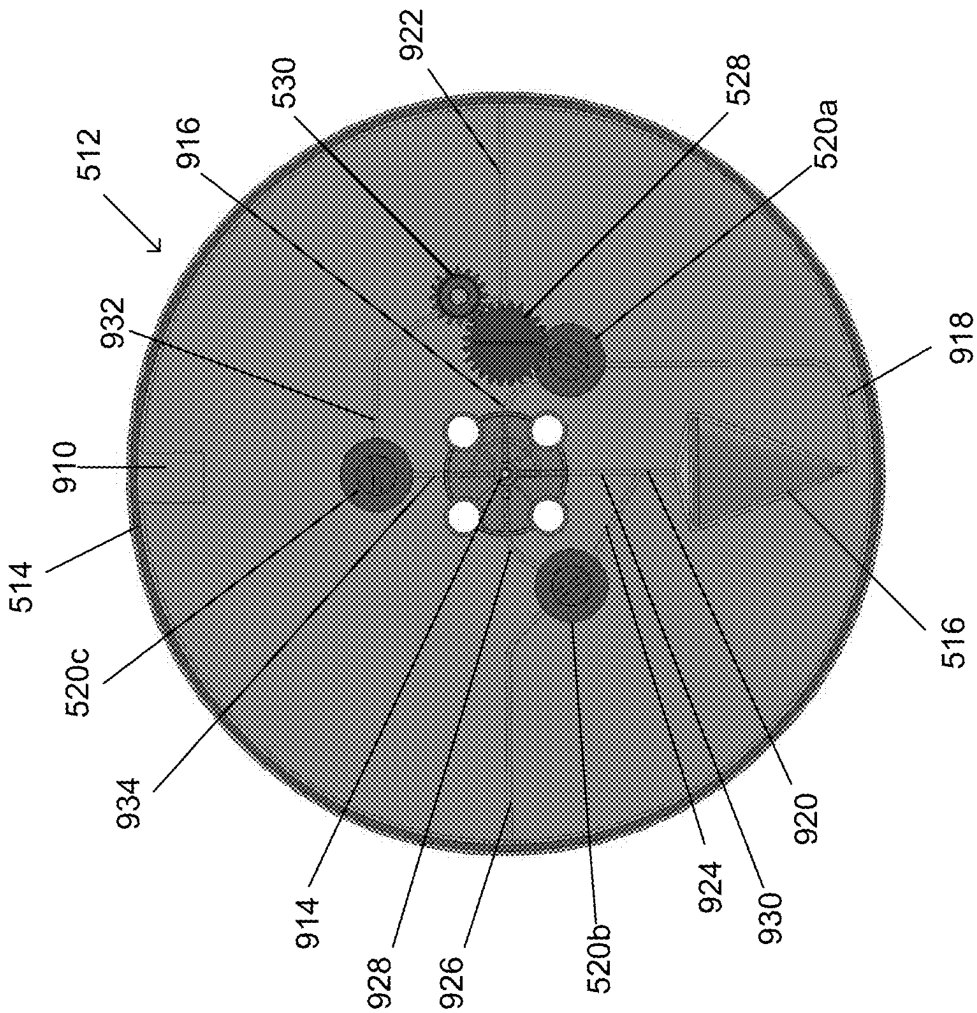

612
520b
522
526a
524a
812a
520a
514
510a
510b
310
516
812b
526b
524b

FIG. 8



922
530
528
520a
512
916
932
918
910
514
520c
516
934
920
914
924
930
928
926
520b

FIG. 9

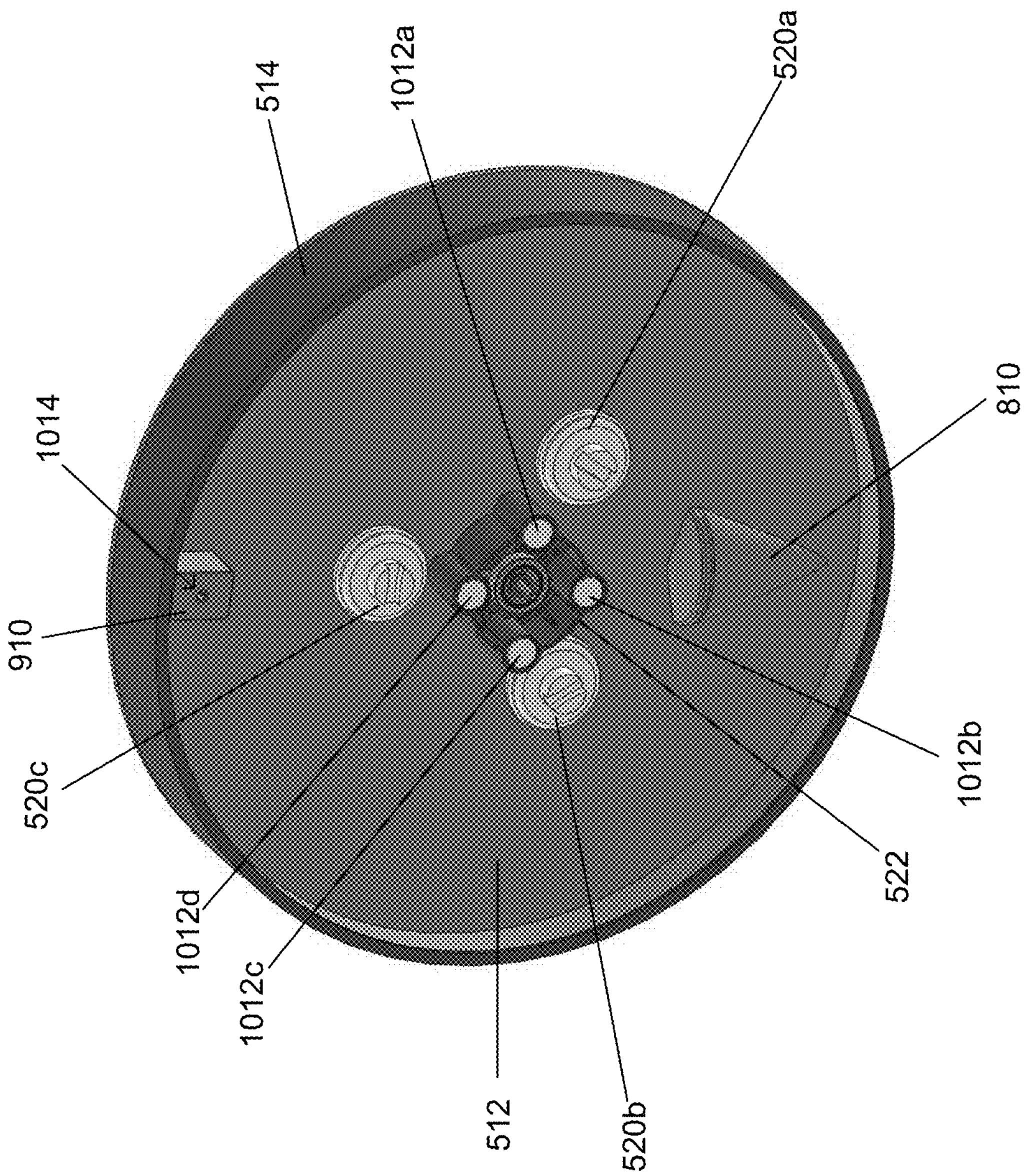

514
1012a
520a
1014
910
520c
810
1012b
522
1012d
1012c
512
520b

FIG. 10

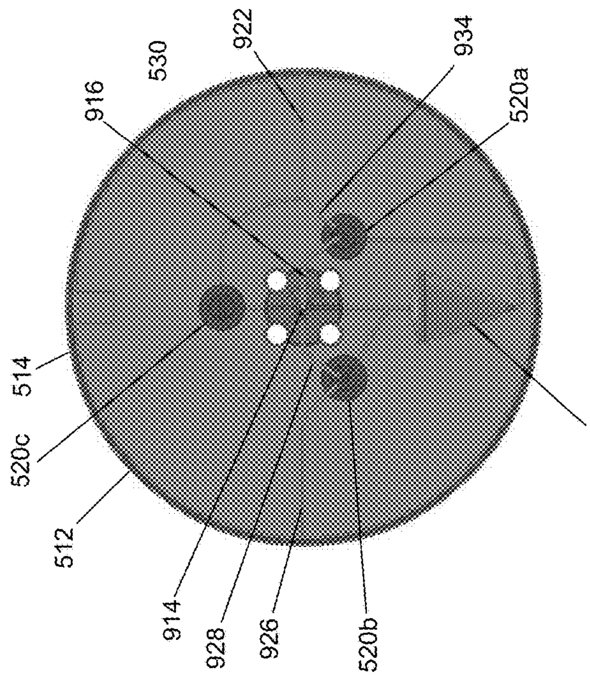


FIG. 11B

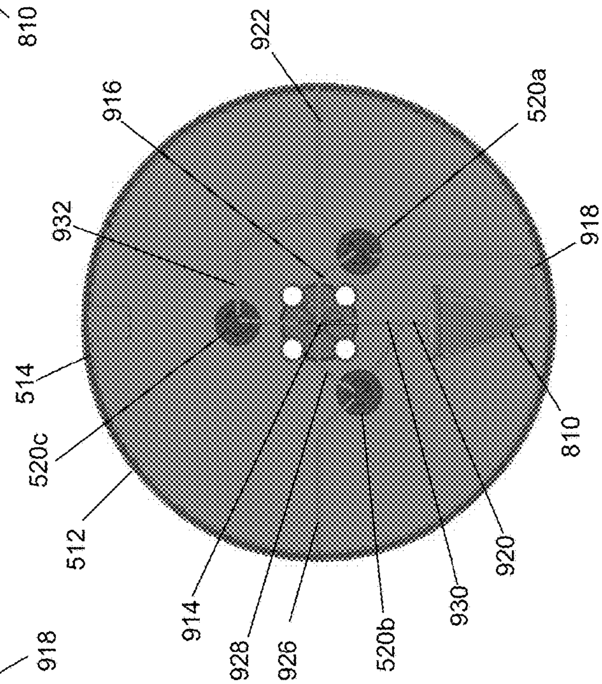


FIG. 11C

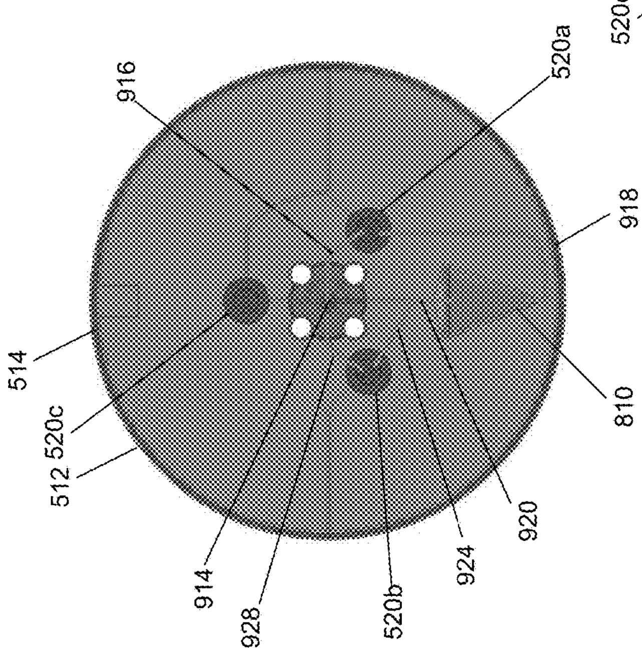


FIG. 11A

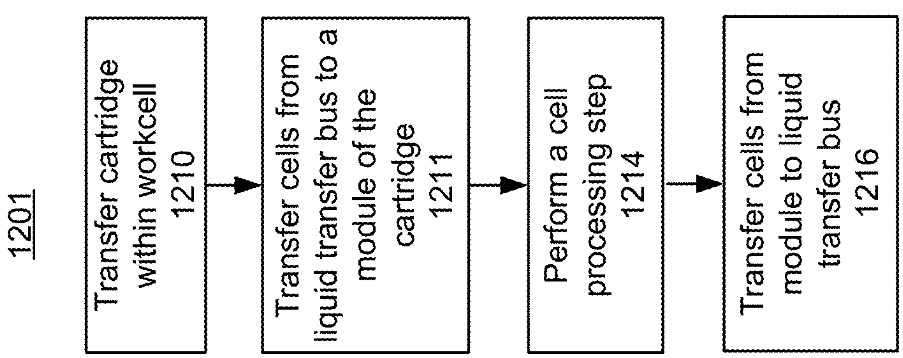


FIG. 12A

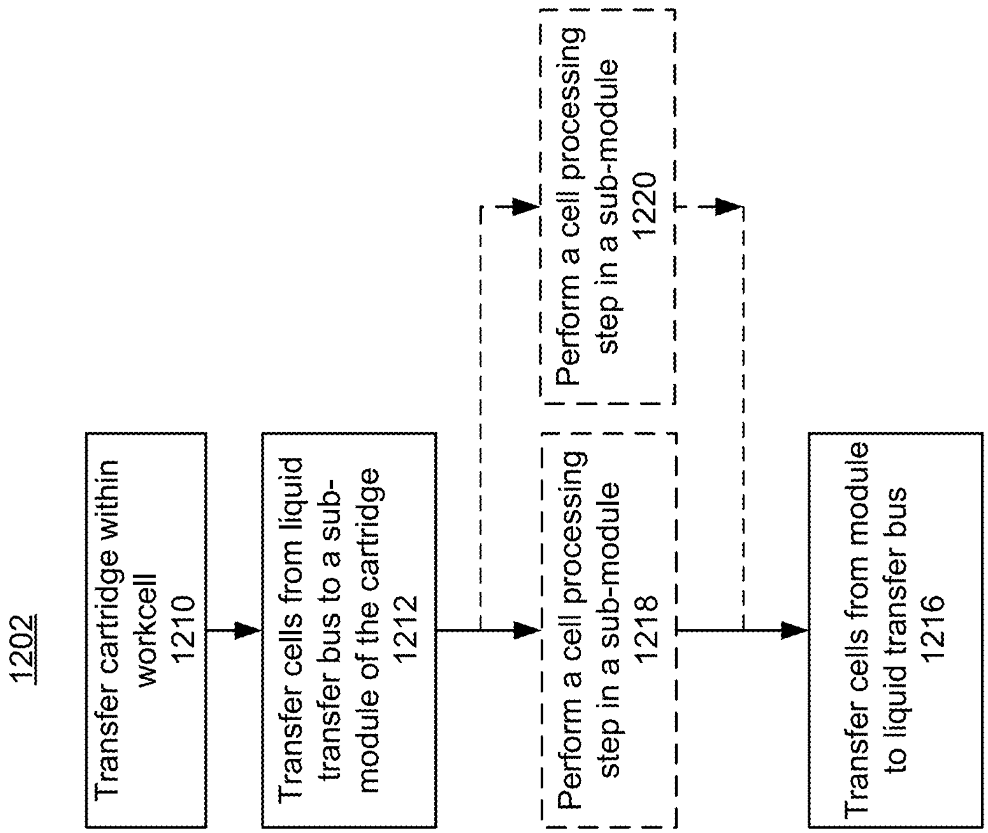


FIG. 12B

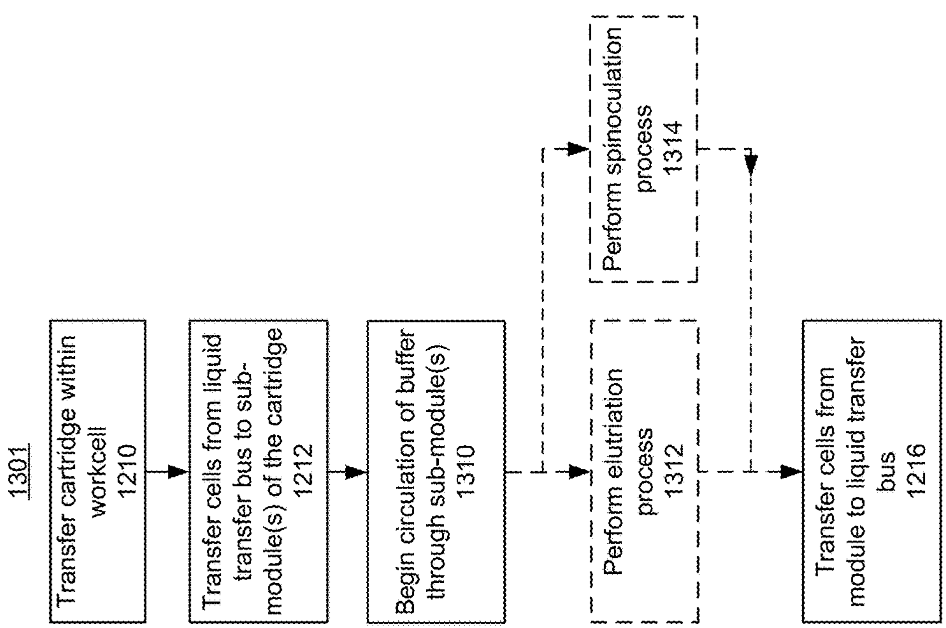


FIG. 13A

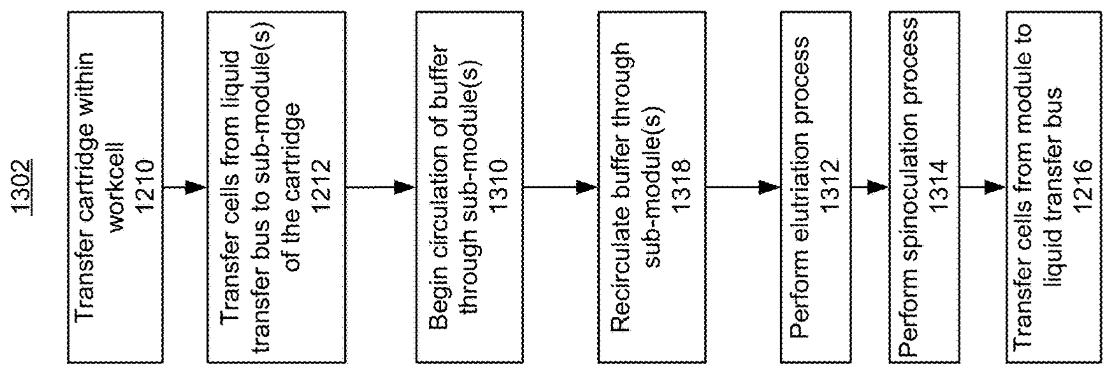


FIG. 13B

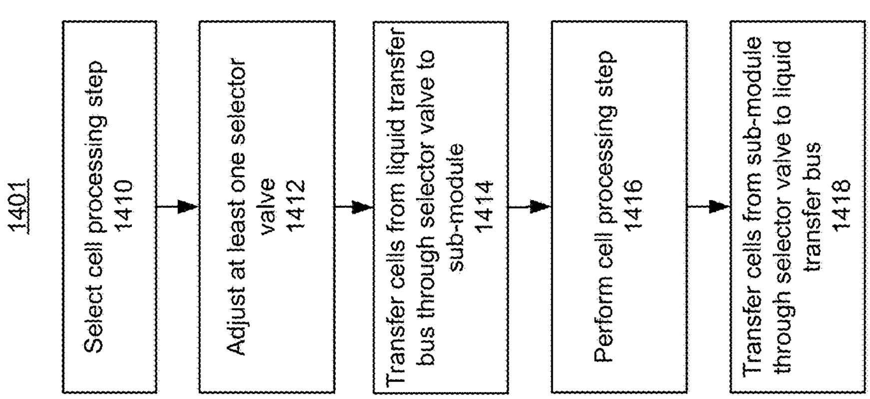


FIG. 14A

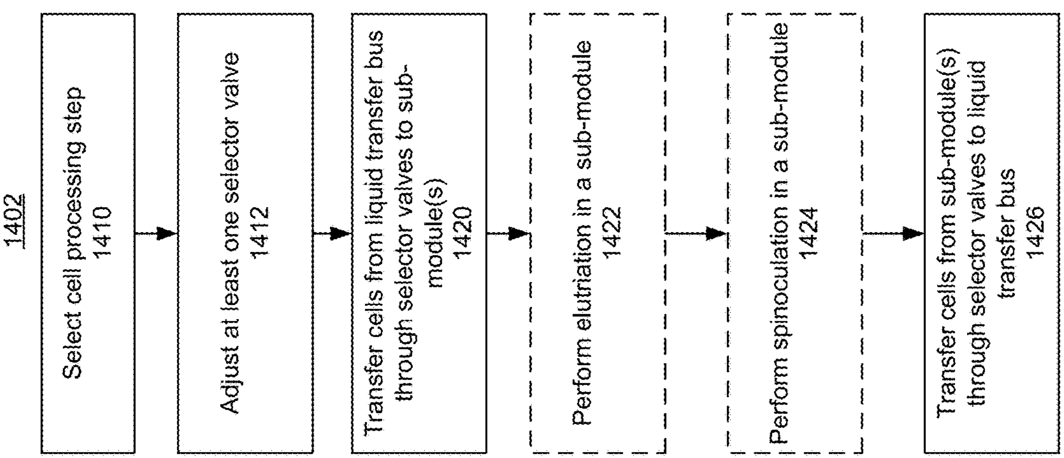


FIG. 14B

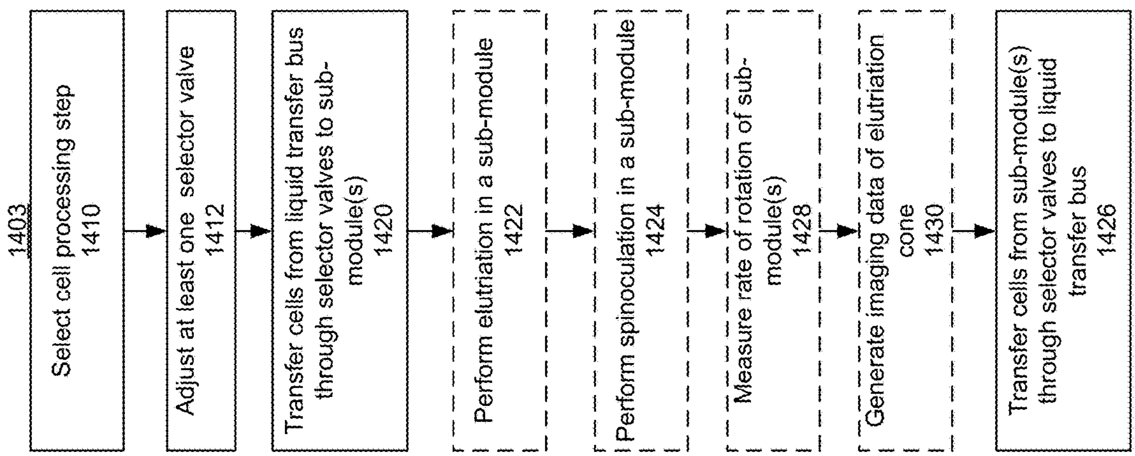


FIG. 14C

SYSTEMS, DEVICES, AND METHODS FOR COMBINED CELL PROCESSES

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims priority to U.S. Provisional Patent Application No. 63/464,386, filed May 5, 2023, the contents of which are hereby incorporated in their entirety by this reference.

TECHNICAL FIELD

The present disclosure relates to systems, devices, and methods for multiple cell processes, for example, spinoculation and elutriation within a cell processing system.

BACKGROUND

Multiple cell processes may be performed on cells during cell-processing. For example, Applicant has developed an elutriation process that separates cells from a fluid based on the size and/or density of the cells and fluid. The elutriation process can include, for example, a counterflow centrifugal elutriation. This elutriation process is typically performed by rotating a fluid container containing the cells at a high rate. Spinoculation on the other hand, binds together cells of different types, such as cells and viral vectors, though the process also requires rotating a fluid container containing the cells at a high rate. Each of these cell processes typically requires separate components, such as motors, fluid containers, and fluid pathways in order to effectively manage the various workflow complexities, such as one process taking longer than another process, or the requirement that one process be completed before another process. Other complexities include the fact that some cell processes require significant volumes of supplementary fluids, which may produce significant waste upon completion of the cell processing. Other cell processes may be preferably performed without the use of supplementary fluids. Additionally, the space required to house the components required to perform the cell processes may be significant. These workflow complexities and housing requirements may be compounded when performing cell processing at a high volume.

Accordingly, additional systems and methods for performing elutriation and spinoculation are desirable, particularly those that may help ameliorate some of the complexities discussed above.

SUMMARY

The present disclosure relates generally to systems, devices, and methods for combined elutriation and spinoculation within an automated cell processing system. In general, the elutriation and spinoculation devices disclosed herein may reside within a cartridge that is used in a cell processing workcell. In some embodiments, the elutriation process comprises a counterflow centrifugal elutriation. In some variations, the cartridge comprises a liquid transfer bus and a module fluidically connected to the liquid transfer bus. The module may comprise two sub-modules, and at least one selector valve configured to direct fluid to at least one of the sub-modules. The two sub-modules may comprise an elutriation sub-module and a spinoculation sub-module. In some variations, at least one selector valve is adjustable to direct fluid to an elutriation chamber of the elutriation sub-module. In some variations, at least one selector valve is adjustable to direct fluid to a spinoculation chamber of the spinoculation sub-module. The at least one selector valve may also be adjustable to direct fluid to an elutriation chamber of the elutriation sub-module and a spinoculation chamber of the spinoculation sub-module in series.

The modules described herein may also comprise a rotor, and in variations where the module has an elutriation sub-module and a spinoculation sub-module, the sub-modules may share the rotor. The module may also comprise a magnetic hub for connecting to a corresponding instrument with a cell processing workcell. The elutriation sub-module and spinoculation sub-module of the module may share the magnetic hub. The elutriation sub-module and spinoculation sub-module may be enclosed within a housing. In some variations, the housing comprises at least one window. In some variations, the elutriation sub-module and spinoculation sub-module share a seal. The seal may comprise a first fluid opening and a second fluid opening.

Methods of elutriation and spinoculation are also described herein. In some variations, the methods comprise transferring a cartridge having cells therein into a cell processing workcell to couple the cartridge to a corresponding instrument of the cell processing workcell, transferring cells from a liquid transfer bus of the cartridge to a module of the cartridge, the module comprising two sub-modules, performing a cell processing step within a sub-module while the cartridge is coupled to the corresponding instrument, and transferring cells from the module to the liquid transfer bus. In some variations, the methods comprise flowing a buffer through at least one sub-module while performing at least one cell processing step. In some variations, the methods comprise measuring the rate of rotation of a rotor of the module using a sensor. In some variations, the methods comprise generating image data of an elutriation sub-module of the module using a sensor.

Additional embodiments, features, and advantages of the invention will be apparent from the following detailed description and through practice of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a block diagram of an illustrative variation of a cell processing system.

FIG. **2**A is a rendering of an illustrative variation of a cell processing system. FIG. **2**B is a rendering of an illustrative variation of a cell processing system.

FIG. **3** is a block diagram of an illustrative variation of a cartridge.

FIG. **4** is a rendering of an illustrative variation of a cartridge.

FIG. **5** is a block diagram of an illustrative module of a cartridge.

FIG. **6**A and FIG. **6**B are renderings of a perspective view of an illustrative module of a cartridge.

FIG. **7**A and FIG. **7**B are renderings of an exploded view of an illustrative module of a cartridge.

FIG. **8** is a rendering of a cross-sectional view of an illustrative module of a cartridge.

FIG. **9** is a rendering of an illustrative rotor of a module.

FIG. **10** is a rendering of a perspective view of an illustrative rotor of a module.

FIG. **11**A, FIG. **11**B, and FIG. **11**C are illustrative variations of fluid flow paths through a rotor of a module.

FIG. **12**A and FIG. **12**B are flowcharts of illustrative variations of cell processing.

FIG. **13**A and FIG. **13**B are flowcharts of illustrative variations of cell processing.

FIG. **14**A, FIG. **14**B, and FIG. **14**C are flowcharts of illustrative variations of cell processing.

DETAILED DESCRIPTION

Devices, systems, and methods for processing cells are described herein. Multiple cell processes, or cell processing steps, may be performed on cells during cell-processing. Elutriation, such as counterflow centrifugal elutriation, separates cells from a fluid based on the size and/or density of the cells and fluid. The elutriation process is typically performed by rotating a fluid container containing the cells at a high rate. The elutriation process may remove red blood cells from an apheresis product. In a typical elutriation process, a buffer may be combined with the apheresis product while spinning the fluid container. The buffer may continuously flow through the fluid container to ensure a seal of the fluid container maintains its efficacy. A spinoculation process binds together cells of different types, such as cells and viral vectors. Spinoculation may enhance the transduction efficiency of viral vectors. The spinoculation process is typically performed by rotating a fluid container containing the cells at a high rate, which causes cells within the fluid to collect along a sidewall of a spinoculation chamber. In a typical spinoculation process, a buffer may not need to continuously flow through the fluid container to perform the spinoculation. However, a fluid may still need to flow through a seal of the fluid container to maintain the seal's efficacy.

Disclosed herein are devices, systems, and methods for performing multiple cell processing steps. The disclosed devices, systems, and methods may be used with a wide range of cell volumes, and in some variations, the devices, systems, and methods disclosed herein utilize multiple fluid modules and/or sensors to assist with accuracy and efficacy.

As described throughout, the cell processing methods, devices, and systems may involve moving a cartridge containing a cell product between a plurality of instruments inside a workcell. One or more instruments may be configured to interface with a cartridge to perform cell processing steps. In some variations, a plurality of cell processing steps may be performed within a single cartridge. For example, a robotic arm may be configured to move a cartridge between instruments, each instrument configured to perform a different cell processing step when coupled to a corresponding module within the cartridge. The cartridge may comprise any number of modules, such as a bioreactor module, a counterflow centrifugal elutriation (CCE) module, a magnetic cell sorter module, an electroporation module, a sorting module (e.g., fluorescence activated cell sorting (FACS) module), a spinoculation module, an acoustic flow cell module, a microfluidic enrichment module, and/or combinations thereof, and the like. In some embodiments, the workcell may process two or more cartridges in parallel. For example, the workcell may comprise a plurality of openings, slots, or bays, wherein each bay is configured to interface with a cartridge, such that multiple bays within the workcell may be in use at any given time. The cell processing systems described herein may reduce operator intervention and increase throughput by automating cartridge movement between instruments using a robot. However, in some embodiments, the cartridge may be moved between instruments manually. Moreover, the automated cell processing system may facilitate sterile liquid transfers between the cartridge and instruments or other components of the system such as a fluid connector (e.g., sterile liquid transfer port), reagent vault, a second cartridge, a sampling vessel e.g., sterile liquid transfer device, combinations thereof, and the like.

I. Cell Processing System

An illustrative cell processing system for use with the devices, systems, and methods is shown in FIG. **1**. Shown there is a block diagram of a cell processing system **100** comprising a workcell **110** and controller **120**. The workcell **110** may comprise one or more of an instrument **112**, a robot **116** (e.g., robotic arm), a reagent vault **118**, a fluid connector **132**, a sterilant source **129**, a fluid source **136**, a pump **138**, and a sensor **140**. A cartridge **114** and a sterile liquid transfer device **142**, part of the cell processing system **100**, may be used within the workcell **110**. The controller **120** may comprise one or more of a processor **122**, a memory **124**, a communication device **126**, an input device **128**, and a display **130**.

The workcell **110** may comprise a fully, or at least partially, enclosed housing inside which one or more cell processing steps are performed in a fully, or at least partially, automated process. In some variations, the workcell may be an open system lacking an enclosure, which may be configured for use in a clean room, a biosafety cabinet, or other sterile location. The cartridge **114** may be moved using the robot **116** to reduce manual labor in the cell processing steps, and sterile liquid transfers into and out of the cartridge may also be performed in a fully or partially automated process. For example, one or more fluids may be stored in a sterile liquid transfer device **142**. In some variations, the sterile liquid transfer device is a portable consumable that may be moved within the system **100**. The sterile liquid transfer devices and fluid connectors described herein may help enable the transfer of fluids in an automated, sterile, and metered manner for automating cell therapy manufacturing.

In some variations, the robot **116** is configured to move cartridges **114** between different instruments to perform a predetermined sequence of cell processing steps. In this way, multiple cartridges **114** may be processed in parallel, as different steps of the cell processing sequence may be performed at the same time on different cartridges.

A fluid connector **132** may be coupled between two or more cartridges **114** to transfer a cell product and/or fluid between the cartridges **114**. Furthermore, a fluid connector **132** may be coupled between any set of fluid-carrying components of the system **100** (e.g., cartridge **114**, reagent vault **118**, fluid source **136**, sterile liquid transfer device **142**, fluid conduit, container, vessel, etc.). For example, a first fluid connector may be coupled between a first cartridge and a sterile liquid transfer device, and a second fluid connector may be coupled between the sterile liquid transfer device and a second cartridge.

Any suitable cell processing may be performed using the systems and devices described herein, and may include steps such as separating, combining, growing, enriching, selecting, sorting, expanding, activating, transducing, electroporating, washing, and the like. In some variations, a method of performing a cell process includes transferring a cartridge containing a cell product to an instrument within a workcell, transferring cells from a liquid transfer bus of the cartridge to a module of the cartridge, performing a cell process step within the module, and transferring cells from the module to the liquid transfer bus.

FIG. **2**A and FIG. **2**B show an illustrative cell processing system for use with the devices, systems, and methods described herein. Shown there is workcell **110**. The workcell may be divided into an interior zone **204** with a feedthrough **206** and quality control (QC) instrumentation **212**. An air filtration inlet (not shown) may provide high-efficiency particulate air (HEPA) filtration to provide ISO7 or better air quality in the interior zone **204**. This air filtration may maintain sterile cell processing in an ISO8 or ISO9 manufacturing environment. The workcell **110** may also have an air filter on the air outlet to preserve the ISO rating of the room. Similar to the workcell described above in reference to FIG. **1**, the workcell **110** may further comprise, inside the interior zone **204**, a bioreactor instrument **214**, a cell separation instrument **216** (e.g., magnetic separation instrument), an electroporation instrument **220**, a counterflow centrifugation elutriation (CCE) instrument **222**, a sterile liquid transfer instrument **224** (e.g., fluid connector), a reagent vault **226**, a spinoculation instrument **230**, and a sterilization system **260**. In other embodiments, different instruments can be combined at one slot or bay, such that two or more instruments can interact with a cartridge positioned in the bay. The reagent vault **226** may be accessible by a user through a sample pickup port **228**. A robot **116** (e.g., support arm, robotic arm) may be configured to move one or more cartridges **114** (e.g., consumables) from any instrument to any other instrument and/or move one or more cartridges **114** to and from a reagent vault. In some embodiments, the workcell **110** may comprise one or more moveable barriers **213** (e.g., access, door) configured to facilitate access to one or more of the instruments in the workcell **110**. In some variations, an outer surface of an enclosure **202** may comprise an input/output device **208** (e.g., display, touchscreen).

In some embodiments, a human operator may load one or more cartridges **114** into the feedthrough **206** via cartridge port **207**. The cartridges **114** may be pre-sterilized, or the feedthrough **206** may sterilize the cartridge **250** using ultraviolet radiation (UV), or chemical sterilizing agents provided as a spray or wash. The feedthrough **206** chamber may optionally be configured to automatically spray, wash, irradiate, or otherwise treat cartridges (e.g., with ethanol and/or isopropyl alcohol solutions) to maintain sterility of the interior zone **204** (e.g., ISO 7 or better). The cartridge **116** may be passed to the biosafety cabinet **206**, where input cell product is provided and loaded to the cartridge using a sterile liquid transfer instrument **224** (e.g., fluid connector) into the cartridge **116**. The user may then move the cartridge **250** back to the feedthrough **206** and initiate automated processing using a computer processor in the computer server rack **210** (e.g., controller **120**). The robot **230** may be configured to move the cartridge **250** in a predefined sequence to a plurality of instruments and stations, with the components of the workcell **200** being controlled by the computer processor of the computer server rack **210**.

Other suitable cell processing systems and aspects thereof are provided e.g., in U.S. patent application Ser. No. 17/198,134, published as U.S. Patent Publication No. 2021/0283565, which is incorporated by reference herein.

A. Workcell

i. Robot

Generally, a robot of the workcell may comprise any mechanical device capable of moving a cartridge from one location to another location within the workcell. For example, the robot may comprise a mechanical manipulator (e.g., an arm) in a fixed location, or attached to a linear rail, or a 2- or 3-dimensional rail system. While shown in some of the figures as being fixed in place or with respect to a rail system, the robot need not be so. For example, in some variations, the robot comprises a wheeled device. Any number of robots may be used within the workcells described herein. For example, in some embodiments, the workcell comprises two or more robots of the same or different type (e.g., two robotic arms each independently configured for moving cartridges between instruments). The robot may also comprise an end effector for precise handling of different cartridges or barcode scanning or radio-frequency identification tag (RFID) reading. The robots for use with the cell processing systems described herein are capable of moving cartridges between slots or bays in the workcell so that the modules within the cartridge can couple to corresponding instruments within the workcell to perform different cell processing steps.

ii. Controller

In embodiments, a cell processing system **100** may comprise a controller **120** (e.g., computing device) comprising one or more of a processor **122**, memory **124**, communication device, **126**, input device **128**, and display **130**. The controller **120** may be configured to control (e.g., operate) the workcell **110**. The controller **120** may comprise a plurality of devices. For example, the workcell **110** may enclose one or more components of the controller **120** (e.g., processor **122**, memory **124**, communication device **126**) while one or more components of the controller **120** may be provided remotely to the workcell **110** (e.g., input device **128**, display **130**).

iii. Processor

The processor (e.g., processor **122**) described here may process data and/or other signals to control one or more components of the system. The processor may be configured to receive, process, compile, compute, store, access, read, write, and/or transmit data and/or other signals. Additionally, or alternatively, the processor may be configured to control one or more components of a device (e.g., console, touchscreen, personal computer, laptop, tablet, server).

In some embodiments, the processor may be configured to access or receive data and/or other signals from one or more of workcell **110**, server, controller **120**, and a storage medium (e.g., memory, flash drive, memory card, database). In some embodiments, the processor may be any suitable processing device configured to run and/or execute a set of instructions or code and may include one or more data processors, image processors, graphics processing units (GPU), physics processing units, digital signal processors (DSP), analog signal processors, mixed-signal processors, machine learning processors, deep learning processors, finite state machines (FSM), compression processors (e.g., data compression to reduce data rate and/or memory requirements), encryption processors (e.g., for secure wireless data transfer), and/or central processing units (CPU). The processor may be, for example, a general-purpose processor, Field Programmable Gate Array (FPGA), an Application Specific Integrated Circuit (ASIC), a processor board, and/or the like. The processor may be configured to run and/or execute application processes and/or other modules, processes and/or functions associated with the system. The underlying device technologies may be provided in a variety of component types (e.g., metal-oxide semiconductor field-effect transistor (MOSFET) technologies like complementary metal-oxide semiconductor (CMOS), bipolar technologies like emitter-coupled logic (ECL), polymer technologies (e.g., silicon-conjugated polymer and metal-conjugated polymer-metal structures), mixed analog and digital, and the like.

The systems, devices, and/or methods described herein may be performed by software (executed on hardware), hardware, or a combination thereof. Hardware modules may include, for example, a general-purpose processor (or microprocessor or microcontroller), a field programmable gate array (FPGA), and/or an application specific integrated circuit (ASIC). Software modules (executed on hardware) may be expressed in a variety of software languages (e.g., computer code), including structured text, typescript, C, C++, C#, Java®, Python, Ruby, Visual Basic®, and/or other object-oriented, procedural, or other programming language and development tools. Examples of computer code include, but are not limited to, micro-code or micro-instructions, machine instructions, such as produced by a compiler, code used to produce a web service, and files containing higher-level instructions that are executed by a computer using an interpreter. Additional examples of computer code include, but are not limited to, control signals, encrypted code, and compressed code.

iv. Memory

The cell processing systems and devices described here may include a memory (e.g., memory **124**) configured to store data and/or information. In some embodiments, the memory may include one or more of a random-access memory (RAM), static RAM (SRAM), dynamic RAM (DRAM), a memory buffer, an erasable programmable read-only memory (EPROM), an electrically erasable read-only memory (EEPROM), a read-only memory (ROM), flash memory, volatile memory, non-volatile memory, combinations thereof, and the like. In some embodiments, the memory may store instructions to cause the processor to execute modules, processes, and/or functions associated with the device, such as image processing, image display, sensor data, data and/or signal transmission, data and/or signal reception, and/or communication. Some embodiments described herein may relate to a computer storage product with a non-transitory computer-readable medium (also may be referred to as a non-transitory processor-readable medium) having instructions or computer code thereon for performing various computer-implemented operations. The computer-readable medium (or processor-readable medium) is non-transitory in the sense that it does not include transitory propagating signals per se (e.g., a propagating electromagnetic wave carrying information on a transmission medium such as space or a cable). The computer code (also may be referred to as code or algorithm) may be those designed and constructed for the specific purpose or purposes. In some embodiments, the memory may be configured to store any received data and/or data generated by the controller and/or workcell. In some embodiments, the memory may be configured to store data temporarily or permanently.

v. Input Device

In some embodiments, an input device **128**, for example, may comprise or be coupled to a display. The input device may be any suitable device that is capable of receiving input from a user, for example, a keyboard, buttons, touch screen, etc. The input device may include at least one switch configured to generate a user input. For example, an input device may include a touch surface for a user to provide input (e.g., finger contact to the touch surface) corresponding to a user input. An input device including a touch surface may be configured to detect contact and movement on the touch surface using any of a plurality of touch sensitivity technologies including capacitive, resistive, infrared, optical imaging, dispersive signal, acoustic pulse recognition, and surface acoustic wave technologies. In embodiments of an input device including at least one switch, a switch may have, for example, at least one of a button (e.g., hard key, soft key), touch surface, keyboard, analog stick (e.g., joystick), directional pad, mouse, trackball, jog dial, step switch, rocker switch, pointer device (e.g., stylus), motion sensor, image sensor, and microphone. A motion sensor may receive user movement data from an optical sensor and classify a user gesture as a user input. A microphone may receive audio data and recognize a user voice as a user input.

In some embodiments, the cell processing system may optionally include one or more output devices in addition to the display, such as, for example, an audio device and haptic device. An audio device may audibly output any system data, alarms, and/or notifications. For example, the audio device may output an audible alarm when a malfunction is detected. In some embodiments, an audio device may include at least one of a speaker, a piezoelectric audio device, a magnetostrictive speaker, and/or a digital speaker. In some embodiments, a user may communicate with other users using the audio device and a communication channel. For example, a user may form an audio communication channel (e.g., VoIP call).

Additionally, or alternatively, the system may include a haptic device configured to provide additional sensory output (e.g., force feedback) to the user. For example, a haptic device may generate a tactile response (e.g., vibration) to confirm user input to an input device (e.g., touch surface). As another example, haptic feedback may notify that user input is overridden by the processor.

vi. Communication Device

In some embodiments, the controller may include a communication device (e.g., communication device **126**) configured to communicate with another controller and one or more databases. The communication device may be configured to connect the controller to another system (e.g., Internet, remote server, database, workcell) by wired or wireless connection. In some embodiments, the system may be in communication with other devices via one or more wired and/or wireless networks. In some embodiments, the communication device may include a radiofrequency receiver, transmitter, and/or optical (e.g., infrared) receiver and transmitter configured to communicate with one or more devices and/or networks. The communication device may communicate by wires and/or wirelessly.

vii. Display

Image data may be output on a display e.g., display **130**) of a cell processing system. In some embodiments, a display may include at least one of a light emitting diode (LED), liquid crystal display (LCD), electroluminescent display (ELD), plasma display panel (PDP), thin film transistor (TFT), organic light emitting diodes (OLED), electronic paper/e-ink display, laser display, and/or holographic display.

viii. Graphical User Interface

In some embodiments, as indicated above, a GUI may be configured for designing a process and monitoring a product. For example, the GUI may be a process design home page. The GUI may indicate that no processes have been selected or loaded. A create icon (e.g., "Create a Process") may be selectable for a user to begin a process design process. In some embodiments, one or more of the GUIs described herein may include a search bar.

B. Cartridge

The cell processing systems described herein may comprise one or more cartridges having one or more modules configured to interface with one or more instruments within the workcell. FIG. **3** shows an illustrative variation of a cartridge **114**. The cartridge **114** may comprise a module **310**, a bioreactor module **318**, an electroporation module **320**, a liquid transfer bus **322**, a magnetic selection module **324**, an auxiliary module **326**, an SLTD tray **328**, a liquid container **330**, and a pump module **332**.

The module **310** may comprise a sub-module **314** and a rotor **312**. The sub-module **314** may be coupled to the rotor **312**. In some embodiments, the module **310** may further comprise a sub-module **316**. The sub-module **316** may be coupled to the rotor **312**. The sub-modules **314** and **316** may share the same rotor **312**. The module **310** may be fluidically connected to the liquid transfer bus **322**. In some variations, the sub-modules **314** and **316** may be fluidically connected to the liquid transfer bus **322** via the module **310**.

The bioreactor module **318** may comprise a vessel configured to culture mammalian cells. Generally, cell and gene therapy products may be grown in a bioreactor to produce a clinical dose which may subsequently be administered to a patient. A number of biological and environmental factors may be controlled to optimize the proliferation speed and success of cell growth.

The electroporation module **320** may be configured to facilitate intracellular delivery of macromolecules (i.e., transfection by electroporation). The electroporation module **320** may contain a continuous flow or batch mode chamber and one or more sets of electrodes for applying direct or alternating current to the chamber. An electrical discharge from one or more capacitors, or current sources, may generate sufficient current in the chamber to promote transfer of a polynucleotide, protein, nucleoprotein complex, or other macromolecule into the cells in the cell product. As with other modules described herein, one or more components used for the process step (here, electroporation) may be provided on the cartridge or in the instrument to which the cartridge interfaces. For example, the capacitor(s) and/or batteries may be provided in the module on the cartridge or in the instrument.

The liquid transfer bus **322** may comprise at least one fluid conduit. The at least one fluid conduit of the liquid transfer bus **322** may be configured to contain at least one fluid. For example, the at least one fluid may be a liquid. In some variations, the at least one fluid may be a gas. In some variations, the at least one fluid may comprise a solution of cells of varying sizes and densities. The liquid transfer bus **322** may comprise at least one fluid inlet and at least one fluid outlet. The liquid transfer bus **322** may comprise at least one valve. The liquid transfer bus **322** may be fluidically connected to at least one module within the cartridge **114**. For example, the liquid transfer bus **322** may be configured to transfer at least one fluid to at least one module within the cartridge **114**. The liquid transfer bus **322** may be fluidically connected to at least one instrument within the workcell **110**. The liquid transfer bus **322** may be in communication with the controller **120**. For example, at least one valve of the liquid transfer bus **322** may open and/or close in response to a command sent by the controller **120**.

The magnetic selection module **324** may perform a magnetic-activated cell selection process. For example, a cell suspension of interest may be immunologically labeled with magnetic particles (e.g., magnetic beads) configured to selectively bind to the surface of the cells of interest. The labeled cells may generate a large magnetic moment when the cell suspension is flowed through a flow cell. The flow cell may be disposed in proximity to a magnet array (e.g., permanent magnets, electromagnet) generating a magnetic field having a gradient across the flow cell to attract the labeled cells for separation, capture, recovery, and/or purification. The magnet array may be configured to generate non-uniform magnetic fields at the edges and the interfaces of the individual magnets so as to cover the full volume of the flow cell such that a magnetophoretic force equals a drag force exerted by the fluid flowing through the flow cell.

The SLTD tray **328** may comprise one or more SLTD slots configured to receive one or more SLTD **142**. In some variations, the SLTD tray **328** may comprise 1 SLTD slot, 2 SLTD slots, 3 SLTD slots, 4 SLTD slots, 5 SLTD slots, 6 SLTD slots, 7 SLTD slots, 8 SLTD slots, 9 SLTD slots, 10 SLTD slots, 11 SLTD slots, 12 SLTD slots, 13 SLTD slots, 14 SLTD slots, or 15 SLTD slots. Each SLTD slot of the SLTD tray **328** may comprise a fluidic conduit configured to fluidically connect with at least one module of the cartridge **114**. For example, each slot of the SLTD tray **328** may be fluidically connected to the liquid transfer bus **322**. In this way, a fluid may flow from the liquid transfer bus **322** to an SLTD **142** contained within one slot of the SLTD tray **328**.

The liquid container **330** may be configured to contain a fluid. In some variations, the fluid is a liquid. In some variations, the fluid is a gas. The gas may be pressurized. The liquid container **330** may be fluidically connected to at least one module of the cartridge **114**. In some variations, the liquid container **330** comprises a plurality of liquid containers. For example, the liquid container **330** may comprise one container, two containers, or three containers. In some variations, the liquid container **330** may be fluidically connected to the liquid transfer bus **322**. In this way, a fluid may flow from the liquid transfer bus **322** to the liquid container **330**.

The pump module **332** may comprise a pump configured to pump fluid in one or more directions along at least one fluid path. For example, the pump module **332** may be configured to pump a fluid to or from one or more of the module **310**, the bioreactor module **318**, the electroporation module **320**, the liquid transfer bus **322**, the magnetic selection module **324**, the auxiliary module **326**, the SLTD tray **328**, and the liquid container **330**.

The auxiliary module **326** may be configured to engage with at least one instrument and/or module. The auxiliary module **326** may comprise at least one electrical connector and/or at least one fluidic connector. In some variations, the auxiliary module **326** may be removed and replaced by any other module.

FIG. **4** shows an illustrative rendering of a cartridge **114**. The cartridge **114** may comprise a module **310**, an electroporation module **320**, a liquid transfer bus **322**, a first magnetic selection module **324*a*** and a second magnetic selection module **324*b***, an auxiliary module **326**, a SLTD tray **328**, a first liquid container **330*a***, a second liquid container **330*b***, a third liquid container **330*c***, and a pump module **332**. In some variations, the cartridge **114** may comprise at least one handle configured for use by a human. In further variations, the cartridge **114** may comprise at least one engagement feature configured for use by a robot. For example, the cartridge **114** may engage with a robot **116** of a workcell **110**.

Various materials may be used to construct the cartridge and the cartridge housing, including metal, plastic, rubber, and/or glass, or combinations thereof. The cartridge, its components, and its housing may be molded, machined, extruded, 3D printed, or any combination thereof. The cartridge may contain components that are commercially available (e.g., tubing, valves, fittings). The commercially available components may be attached or integrated with custom components or devices. The housing of the cartridge may constitute an additional layer of enclosure that further protects the sterility of the cell product.

In some embodiments, the modules may be integrated in a fixed configuration within the cartridge or a housing within the cartridge. Additionally, or alternatively, the modules may be configurable or moveable within the cartridge, permitting various formats of cartridges to be assembled. For example, the cartridge can be a single, closed unit with fixed components for each module, or the cartridge may contain configurable modules coupled by configurable fluidic, mechanical, optical, and electrical connections. In some variations, one or more sub-cartridges, each containing a set of modules, or one or more sub-modules within a single cartridge, may be used to perform various cell processing workflows. The modules may each be provided in a distinct housing or may be integrated into a single housing, cartridge, or sub-cartridge with other modules. In some embodiments, multiple cartridges may be used to process a single cell product through transfer of the cell product from one cartridge to another cartridge of the same or different type and/or by splitting cell product into more cartridges and/or pooling multiple cell products into fewer cartridges.

Generally, each of the instruments within the workcell interfaces with its respective module or modules on the cartridge. For example, when a cartridge has an electroporation module, it may be moved by the robot to the electroporation instrument within the workcell to perform electroporation on the cells within the cartridge. One advantage of such split module/instrument designs is that expensive components (e.g., motors, sensors, heaters, lasers, etc.) may be retained in the instruments of the system while less expensive components reside in the cartridge, which is typically disposable and configured for single-use. The use of disposable cartridges may eliminate the need to sterilize cartridges between use. Furthermore, having multiple instruments within the workcell further helps allow for the parallel utilization of those instruments when multiple cartridges are used within the workcell. In contrast, most conventional semi-automated instruments have instrument components that sit idle and are incapable of simultaneous parallel use.

C. Combined Spinoculation and CCE Module

As described above, the cell processing steps that may be performed include spinoculation and counterflow centrifugal elutriation. In some embodiments, where spinoculation is performed, the cartridge comprises a spinoculation sub-module and the workcell comprises a corresponding spinoculation instrument. In some embodiments, where counterflow centrifugal elutriation is performed, the cartridge comprises an elutriation sub-module and the workcell comprises a corresponding elutriation instrument. In some embodiments, the cartridge comprises both of a spinoculation sub-module and an elutriation sub-module, and the workcell comprises an instrument corresponding to both spinoculation and elutriation. In some embodiments, spinoculation and counterflow centrifugal elutriation may be performed in series.

FIG. **5** provides a schematic illustration of a workcell **110** comprising an exemplary cartridge **114**. The cartridge **114** may comprise a module **310**. The module **310** may be coupled to one or more other modules of the cartridge **114**. In some variations, the module **310** may be fluidically connected to a liquid transfer bus **322** via at least one fluid conduit. For example, liquid transfer bus **322** may comprise at least one valve configured to begin and/or stop fluid flow through the at least one fluid conduit coupled to the module **310**. In some variations, the liquid transfer bus **322** may comprise a first fluid conduit configured to direct fluid to the module **310** and a second fluid conduit configured to receive fluid from the module **310**. In some variations, there may be a plurality of fluid conduits directing fluid to the module **310** and a plurality of fluid conduits receiving fluid from the module **310**. In some variations, the module **310** may be fluidically connected to any other module of the cartridge **114** such as the pump module **332**, the liquid container **330**, and/or the bioreactor module **318**. For example, the module **310** may receive fluid from the liquid transfer bus **322** and output fluid to the liquid container **330**, or vice versa. In some embodiments, the module **310** may receive fluid from and/or transmit fluid to any instrument within the workcell **110**. For example, the module **310** may receive fluid from and/or transmit fluid to an elutriation instrument and/or a spinoculation instrument.

The module **310** may be mechanically coupled to one or more modules of the cartridge **114**. For example, the module **310** may be fastened to a magnetic selection module **324**, pump module **322**, auxiliary module **326**, or any other module of the cartridge **114** via at least one screw, glue, adhesive, tape, clamp, or any other suitable means. In some variations, the module **310** may be mechanically coupled to a sidewall of the cartridge **114**. In further variations, the module **310** may be positioned within the cartridge **114** via a friction force and/or a compressive force. The module **310** may comprise metal, plastic, rubber, glass, or combinations thereof. The material used to construct the module **310** may be selected based on compatibility with one or more modules within the cartridge **114**, one or more instruments within the workcell **110**, and/or one or more fluids used in the workcell **110**.

The module **310** may comprise a container or housing configured to securely contain at least one component. In some embodiments, the module **310** may comprise a housing **510**. The housing **510** may comprise a sub-module **314**. In some variations, the housing **510** may comprise a sub-module **316**, which may be different than the sub-module **314**. In further variations, the sub-modules **314** and **316** may be substantially identical. In some variations, the sub-module **314** may comprise a spinoculation sub-module. The sub-module **314** may further comprise a spinoculation chamber **514**. The sub-module **314** may be configured to perform a spinoculation process. In some variations, the sub-module **316** may comprise an elutriation sub-module. The sub-module **316** may further comprise an elutriation chamber **516** comprising an elutriation cone. The sub-module **316** may be configured to perform an elutriation process.

The housing **510** may further comprise at least one component configured to rotate. In some embodiments, the housing **510** may comprise a rotor **512**. The rotor **512** may be coupled to the sub-module **314**. In some variations, the rotor **512** may also be coupled to the sub-module **316**. In this way, the sub-modules **314** and **316** may advantageously share the same rotor **512** to alleviate the workflow challenges and housing complexities described above. The rotor **512** may be configured to rotate at a high rate. For example, the rotor may rotate at about 500 rpm, about 1000 rpm, about 2000 rpm, about 3000 rpm, about 4000 rpm, about 5000 rpm, or about 6000 rpm. In some variations, the rate of rotation may be determined by the cell processing step to be performed. In further variations, the rate of rotation may be determined by the volume of fluid contained within the sub-module **314** and/or sub-module **316**. The rotor **512** may be configured to rotate within the housing **510** without damaging any other component within the housing **510**, module **310**, cartridge **114**, and/or workcell **110**. The rate of rotation of the rotor **512** may be determined by a vibration level associated with the housing **510** and/or module **310** while the rotor **512** rotates.

The rotor **512** may be configured to direct fluid therethrough. In some embodiments, the rotor **512** may comprise a selector valve **520**. The selector valve **520** may comprise a rotary valve. The selector valve **520** may comprise at least one fluidic path therethrough. The selector valve **520** may be configured to rotate between one or more positions. In some variations, the rotation of the selector valve **520** may adjust the fluidic path therethrough. The selector valve **520** may comprise a first selector valve **520***a*. In some variations, the selector valve **520** may further comprise a second selector valve **520***b* and a third selector valve **520***c*. In some variations, one or more of the selector valves **520***a*, **520***b*, **520***c* may advantageously be fluidically connected to one or more of the sub-modules **314**, **316**. For example, the selector valves **520***a*, **520***b*, **520***c* may define a first fluid flow path configured to direct fluid to the sub-module **314**. The first fluid flow path may comprise a fluidic connection between the liquid transfer bus **322** and the spinoculation chamber **514**. In another example, the selector valves **520***a*, **520***b*, **520***c* may define a second fluid flow path configured to direct fluid to the sub-module **316**. The second flow path may comprise a fluidic connection between the liquid transfer bus **322** and the elutriation chamber **516**. In yet another example, the selector valves **520***a*, **520***b*, **520***c* may define a third fluid flow path to direct fluid to each of the sub-modules **314** and **316** in series. The third flow path may comprise a fluidic connection between the liquid transfer bus **322** and each of the spinoculation chamber **514** and elutriation chamber **516**. The selector valves **520***a*, **520***b*, **520***c* can thus advantageously enable controlled operation of either one or both of the spinoculation chamber **514** or elutriation chamber **516**.

The housing **510** may further comprise at least one mechanical means for adjusting a configuration of the selector valve **520**. In some embodiments, the housing **510** may comprise a gear **528**. The gear **528** may define an axis of rotation. The gear **528** may comprise a plurality of gear teeth. The gear **528** may comprise a thickness determined by an overall thickness of the housing **510**. For example, the gear **528** may be configured to sit flush with an external surface of the housing **510**. The gear **528** may be configured to engage with at least one selector valve **520**. For example, teeth of the gear **528** may correspond to and engage with the first selector valve **520***a*. The gear **528** may be configured to adjust a position of the selector valve **520**. For example, rotation of the gear **528** may adjust a position of one or more of the selector valves **520***a*, **520***b*, **520***c*. In some variations, adjusting the selector valve **520** may comprise rotating the selector valve **520**.

The selector valve **520** may comprise a first position and a second position. The selector valve **520** may be adjusted between the first and second positions via rotation in a clockwise and/or a counterclockwise direction. The selector valve **520** may further comprise a detent. The detent may comprise a protrusion and/or depression configured to apply a force to the selector valve **520**. The force applied by the detent may prevent rotation of the selector valve **520** under passive conditions. For example, the force applied to the detent may be greater than the force of gravity acting on the selector valve **520**. The force applied by the detent may be overcome by rotation of the gear **528** that, in turn, adjusts the position of the selector valve **520**. The detent may comprise a first detent and a second detent. The first and second detents may be located adjacent to the outer circumference of the selector valve **520** such that the detent is engageable when the selector valve **520** rotates. In this way, the detent may be located within the path of rotation of the selector valve **520**. In some variations, the first detent is in a first location along the path of rotation and the second detent is in a second location along the path of rotation. In some variations, the first location of the first detent is separated by approximately 180 degrees from the second position of the second detent. The first detent may be associated with the first position of the selector valve **520** and the second detent may be associated with the second position of the selector valve **520**.

The rotor **512** may further comprise a hard stop associated with the selector valve **520**. The hard stop may comprise a protrusion configured to apply a force to the selector valve **520**. The force applied by the hard stop may be configured to prevent rotation of the selector valve **520**. The force applied by the hard stop may not be overcome by rotation of the gear **528**. In some variations, each of the selector valves **520***a*, **520***b*, **520***c* have separate hard stops.

The workcell **110** may be configured to engage with the gear **528** of the housing **510**. In some embodiments, the workcell **110** may further comprise a gear **530**. The gear **530** may be associated with an instrument of the workcell **110**. For example, the gear **530** may be associated with one or more an elutriation instrument and a spinoculation instrument. The gear **530** may be in communication with the controller **120**. For example, a user may input a command into the controller **120** to rotate the gear **530**. The gear **530** may be configured to engage with the gear **528**. For example, the gear **530** may rotate in response to a command transmitted by the controller **120**, which in turn rotates the gear **528**. In this way, a user may adjust the selector valve **520** via the gears **528**, **530**.

The housing **510** may be configured to rotate one or more components contained therein. In some embodiments, the housing **510** further comprises a magnetic hub **522**. The magnetic hub **522** may be coupled to the rotor **512**. The magnetic hub **522** may be configured to rotate the rotor **512** about a rotation axis defined by the magnetic hub **522**. The magnetic hub **522** may comprise one or more magnets. For example, the magnetic hub **522** may comprise one or more electromagnets. The magnetic hub **522** may be electrically connected to the cartridge **114**. For example, the magnetic hub **522** may rotate in response to an electrical signal sent by the cartridge **114** to one or more magnets within the magnetic hub **522**. In some variations, the rate of rotation of the rotor **512** may be determined by the amplitude of the electrical signal sent to the magnetic hub **522**. In some variations, the magnetic hub **522** may be magnetically coupled to the cartridge **114**. For example, the cartridge **114** may comprise at least one magnet that corresponds to the magnetic hub **522**. The magnetic hub **522** may be configured to rotate in response to a command from the controller **120**.

The housing **510** may further comprise at least one component configured to fluidically communicate with one or more modules of the cartridge **114** and/or the workcell **110**. In some embodiments, the housing **510** comprises a seal **524** and a fluid opening **526**. The fluid opening **526** may be configured to receive at least one fluid. The fluid opening **526** may be fluidically connected to the liquid transfer bus **322**. In some variations, the fluid opening **526** may be fluidically connected to one or more of the bioreactor module **318**, auxiliary module **326**, liquid container **330**, magnetic selection module **324**, and pump module **332**. The fluid opening **526** may be fluidically connected to the rotor **512**. For example, the liquid transfer bus **322** may direct fluid through the fluid opening **524** and to one or more fluid paths of the rotor **512**. In a further example, the rotor **512** may direct fluid through the fluid opening **524** and to the liquid transfer bus **322**. In some variations, the fluid opening **524** comprises a stainless steel tube permanently molded into the rotor **512**. The seal **524** may be configured to fluidically seal a fluid path through the fluid opening **526**.

The seal **524** may be referred to as a dynamic seal. The seal **524** may be configured to be in direct contact with at least one fluid. The seal **524** may maintain its efficacy as a fluid seal when the seal **524** is substantially wet. In some variations, the seal **524** may require continuous fluid flow therethrough to maintain its efficacy as a fluid seal. In further variations, the seal **524** may not require continuous fluid flow therethrough to maintain its efficacy as a fluid seal. In some variations, the seal **524** may comprise an O-ring. In some variations, the seal **524** may comprise one or more of nitrile, neoprene, ethylene propylene, silichamber, fluorocarbon, and polytetrafluoroethylene (PTFE).

The housing **510** may further comprise at least one feature configured to provide means to measure a parameter associated with the conditions within the housing **510**. In some embodiments, the housing **510** may comprise a window **532**. The window **532** may comprise a substantially transparent material in a wall of the housing **510**. In some variations, the window **532** may comprise one or more of polyethylene terephthalate, polypropylene, polycarbonate, glass, and other transparent material. The window **532** may comprise a circle, a rectangle, a triangle, a trapezoid, or any other geometric shape. The window **532** may be operatively coupled with a sensor **140** of the workcell **110**. For example, a sensor **140** comprising an optical sensor may be configured to measure at least one parameter of the housing **510** through the window **532**. In some variations, the sensor **140** may be configured to measure a rate of rotation of the housing **510**. In further variations, the sensor **140** may be configured to measure a parameter associated with the elutriation chamber **516**. For example, the sensor **140** may be configured to measure one or more of a quantity and a position of cells within the elutriation chamber **516**. In some variations, at least a portion of a wall of the elutriation chamber **516** comprises a transparent material. In yet further variations, the sensor **140** may be configured to measure a parameter associated with the spinoculation chamber **514**. For example, the sensor **140** may be configured to measure one or more of a quantity and a position of cells within the spinoculation chamber **514**. In some variations, the sensor **140** comprises a first sensor **140***a* and a second sensor **140***b*.

FIG. **6**A and FIG. **6**B show renderings of an exemplary embodiment of a module **310**. The module **310** may comprise a first housing **510***a* and a second housing **510***b*. The first housing **510***a* and second housing **510***b* may be coupled together. In some variations, the housings **510***a*, **510***b* may be coupled together via at least one screw, an adhesive, a weld, or any means suitable to maintain a rigid connection. In further variations, the housings **510***a*, **510***b* may be removably coupled. For example, the housings **510***a*, **510***b* may be separated to allow a user to conduct maintenance of any component contained therein. In some variations, the housings **510***a*, **510***b* may comprise a honeycomb structure. The honeycomb structure may be desired to reduce the mass of the housings **510***a*, **510***b* while maintaining sufficient rigidity such that the housing **510***a*, **510***b* may withstand the forces associated with rotating at high rates.

The housing **510***a* may further comprise a gear **528** and a groove **616**. The gear **528** may be adjacent to the groove **616**. The groove **616** may be configured to receive the gear **530** of the workcell **110**. The groove **616** may comprise a depth associated with a thickness of the gear **530**. For example, the depth of the groove **616** may enable the gear teeth of the gear **530** to fully engage with the gear teeth of the gear **528**. In some variations, the depth of the groove **616** may prevent the gear **530** from contacting the external surface of the housing **510***a* while the gear **530** rotates. In some variations, a length of groove **616** may be sufficient to allow the gear **530** to fully disengage from the gear **528**. For example, the gear **530** may be engaged with the gear **528** when in a first position within the groove **616** and disengaged from the gear **528** when in a second position within the groove **616**. In some embodiments, the gear **530** may be in the second position when the rotor **512** rotates. In this way, the gear **530** may not rotate with the rotor **512**. In some variations, a width of the groove **616** may be determined by the diameter of the gear **530**. In some embodiments, the diameter of the gear **530** is less than the diameter of the gear **528**. In some variations, the diameter of the gear **530** is greater than the diameter of the gear **528**.

The housing **510***a* may be configured to engage with a plurality of sensors. In some embodiments, the housing **510** further comprises a first window **532***a* and a second window **532***b*. Each of the windows **532***a*, **532***b* may comprise a substantially transparent material in a wall of the housing **510***a*. Each of the windows **532***a*, **532***b* may be operatively coupled with at least one sensor. For example, the window **532***a* may be operatively coupled to a sensor **140***a*. In another example, the window **532***b* may be operatively coupled to a sensor **140***b*.

The housing **510***a* may comprise at least one removable component. In some embodiments, the housing **510***a* further comprises a cover **610**. The cover **610** may be configured to enclose at least one fluid path defined by the housing **510***a*. The cover **610** may be removable without otherwise removing the module **310** from the cartridge **114**. The cover **610** may comprise a fluid outlet. The fluid outlet may comprise a through hole. In some variations, the fluid outlet may be fluidically connected to one or more instruments within the workcell **110**. In further variations, the fluid outlet is fluidically connected to the liquid transfer bus **322**. In yet further variations, the fluid outlet is fluidically sealed to prevent fluid flow therethrough.

The housing **510***b* may comprise a fluid opening **526** and a connector **612**. The fluid opening **526** may be fluidically connected to at least one module of the cartridge **114**. For example, the fluid opening **526** may be fluidically connected to the liquid transfer bus **322**. The connector **612** may comprise a mechanical fastener configured to engage with at least one module of the cartridge **114**. For example, the connector **612** may comprise a protrusion. In some variations, there may be a corresponding hole in a wall of the cartridge **114**. For example, the connector **612** may be configured to secure the module **310** in a position within the cartridge **114**. In further variations, there may be a corresponding hole in any other module within the cartridge **114**. In this way, the housing **510***b* may be mechanically fastened to one or more modules.

FIG. **7**A and FIG. **7**B provides an exploded view of an exemplary embodiment of the module **310**. The first housing **510***a* may be coupled to the second housing **510***b*. The housings **510***a*, **510** may comprise a substantially rectangular outer perimeter. Between the first and second housings **510***a*, **510***b* may be a magnetic hub **522**. The magnetic hub **522** may be coupled to one or both of the housings **510***a*, **510***b*. In some variations, the magnetic hub **522** may be configured to rotate freely relative to the housings **510***a*, **510***b*. The module **310** may further comprise a rotor **512**. The rotor **512** may be coupled to one or both of the housings **510***a*, **510***b*. In some variations, the rotor **512** may be configured to rotate freely relative to the housings **510***a*, **510***b*. The rotor **512** may comprise an opening configured to receive the magnetic hub **522**. The magnetic hub **522** may protrude through the rotor **522**.

The module **310** may further comprise components associated with a spinoculation process. In some embodiments, the module **310** comprises a spinoculation chamber **514**. The spinoculation chamber **514** may comprise an outer chamber wall **514***a* and an inner chamber wall **514***b*. The chamber walls **514***a*, **514***b* may each comprise a circular shape. In some variations, there may be a gap between the chamber walls **514***a*, **514***b*. The size of the gap may determine a volume of fluid contained therein. For example, the gap may comprise about 0.5 mm, 1 mm, 2 mm, 3 mm, 4 mm, 5 mm, 6 mm, 7 mm, 8 mm, 9 mm, or 10 mm. The module **310** may further comprise a seal **710**. The seal **710** may form a fluid-tight seal between the outer chamber wall **514***a* and inner chamber wall **514***b*. The seal **710** may also form a fluid-tight seal between the spinoculation chamber **514** and the second housing **510***b*. In some variations, the seal **710** may comprise a rubber O-ring.

FIG. **8** shows a cross-sectional view of an exemplary embodiment of the cartridge **310**. The cartridge **310** may comprise a first housing **510***a*, a second housing **510***b*, a first selector valve **520***a*, a second selector valve **520***b*, a third selector valve (not shown), a magnetic hub **522**, and a connector **612**. The cartridge **310** may further comprise an elutriation chamber **516**. The elutriation chamber **516** may be defined by the rotor **512**. In some variations, the elutriation chamber **516** may be defined by one or more of the housings **510***a*, **510***b*.

The module **310** may further comprise a first fluid opening **526***a* through the first housing **510***a* and a second fluid opening **526***b* through the second housing **510***b*. The first fluid opening **526***a* may comprise a tube with a lumen therethrough, the tube extending from the first housing **510***a*. The second fluid opening **526***b* may comprise a tube with a lumen therethrough, the tube extending from the first housing **510***b*. Surrounding each fluid openings **526***a*, **526***b* may be a seal **524**. For example, a first seal **524***a* may surround the first fluid opening **526***a*. Similarly, a second seal **524***b* may surround the second fluid opening **526***b*. The first and second seals **524***a*, **524***b* may be configured to prevent fluid leakage from a fluid path through the fluid openings **526***a*, **526***b* into other volumes within the module **310**.

The magnetic hub **522** may further comprise a bearing **812**. The bearing **812** may comprise a first bearing **812***a* and a second bearing **812***b*. The first bearing **812***a* may be contained within first housing **510***a*. The second bearing **812***b* may be contained within the second housing **510***b*. The first and second bearings **812***a*, **812***b* may facilitate the rotation of one or more of the magnetic hub **522** and the rotor **512**. The bearing **812** may be shared by the sub-modules **314**, **316**.

FIG. **9** shows an exemplary embodiment of a rotor **512**. The rotor **512** may comprise a first selector valve **510***a*, a second selector valve **510***b*, and a third selector valve **510***c*. In some variations, the selector valves **510***a*, **510***b*, **510***c* are positioned approximately equidistant apart. The rotor **512** may further comprise a plurality of fluid paths. The plurality of fluid paths may be configured to direct fluid through at least one selector valve. A fluid may flow into the rotor through a fluid opening **914**. The fluid opening **914** may be fluidically connected to each of the selector valves **520***a*, **520***b*, **520***c* through at least one fluid flow path. For example, the fluid opening **914** may be fluidically connected to the first selector valve **520***a* via flow path **916**. The fluid opening **914** may be fluidically connected to the second selector valve **520***b* via flow path **928**. The fluid opening **914** may be fluidically coupled to the third selector valve **520***c* via flow path **934**. The fluid may flow through any of the flow paths **916**, **928**, **934** based on the configuration of each selector valve. For example, each selector valve may comprise a clockwise (CW) position and a counterclockwise (CCW) position. In some variations, the clockwise position may be a first position and the counterclockwise position may be a second position. Fluid may or may not flow through a selector valve based on its position. Each selector valve may be positioned independently of any other selector valve. In some variations, each selector valve may be positioned in conjunction with at least one other selector valve.

The rotor **512** may be coupled to the elutriation chamber **516**. The elutriation chamber **516** may be fluidically connected to at least one selector valve **520**. For example, the flow path **918** may fluidically connect the elutriation chamber **516** to the selector valve **520***a*. Similarly, the flow path **920** may fluidically connect the elutriation chamber **516** to the selector valve **520***b*. In yet another example, the flow path **930** may fluidically connect the elutriation chamber **516** to the selector valve **520***c*.

Additionally or separately, the rotor **512** may be coupled to a spinoculation chamber **514**. The spinoculation chamber **514** may be fluidically connected to at least one selector valve **520**. For example, the flow path **922** may fluidically connect the spinoculation chamber **514** to the selector valve **520***a*. Similarly, the flow path **926** may fluidically connect the spinoculation chamber **514** to the selector valve **520***b*. In yet another example, the flow path **932** may fluidically connect the spinoculation chamber **514** to the selector valve **520***c*.

The rotor **512** may be coupled to a gear **528**. For example, the gear **528** may be operatively coupled to the first selector valve **520***a*. The gear **528** may comprise a plurality of teeth configured to engage with corresponding teeth on a gear **530**. The gear **530** may be coupled to the cartridge **114**. The gear **530** may be adjusted based on an input from the controller **130**. The gear **530** may rotate in either a clockwise or counterclockwise direction. In some variations, the gear **530** may be engaged with the gear **528** such that rotational movement of the gear **530** may cause the gear **528** to rotate. The gear **528** may rotate in either a clockwise or counterclockwise direction. Subsequently, the rotation of gear **528** may rotate the first selector valve **520***a*. In some embodiments, the rotation of gear **528** may also rotate or adjust the second selector valve **520***b* and/or the third selector valve **520***c*. In some variations, the gear **528** may also be manually rotated by, for example, a user. The selector valves **520***a*, **520***b*, **520***c* may also be manually rotated or adjusted by, for example, a user.

The rotor **512** may further comprise a protrusion **910**. The protrusion **910** may extend from a surface of the rotor **512**. The protrusion **910** may comprise at least one planar surface configured to receive at least one optical feature. For example, the optical feature may comprise reflective tape configured to be detected by an optical sensor. In some variations, the optical feature may comprise a hole, an indent, a depression, a section of color that contrasts with a surrounding color, or any other feature configured to be measured by an optical sensor.

FIG. **10** provides a rendering of an illustrative embodiment of the rotor **512**, elutriation chamber **516**, and spinoculation chamber **514**. The rotor **512** may further comprise a magnetic hub **522** comprising at least one magnet **1012**. For example, the magnet **1012** may comprise a first magnet **1012***a*, a second magnet **1012***b*, a third magnet **1012***c*, and a fourth magnet **1012***d*. In some variations, the magnets **1012***a*, **1012***b*, **1012***c*, **1012***d* may comprise electromagnets. The magnets **1012***a*, **1012***b*, **1012***c*, **1012***d* may each receive an electrical signal sent by the controller **120**. In some variations, the electrical signal received by each of the magnets **1012***a*, **1012***b*, **1012***c*, **1012***d* is substantially equivalent. In further variations, the electrical signals received by each of the magnets **1012***a*, **1012***b*, **1012***c*, **1012***d* is different. The rate of rotation of the rotor **512** may be controlled by the amplitude of the electrical signals transmitted to the magnets **1012***a*, **1012***b*, **1012***c*, **1012***d*. In some variations, the magnets **1012***a*, **1012***b*, **1012***c*, **1012***d* may comprise permanent magnets.

FIG. **10** provides an exemplary embodiment of the protrusion **910**. The protrusion **910** may comprise an optical feature **1014** coupled to at least one surface of the protrusion **910**. In some variations, the optical feature **1014** may comprise a feature configured to be measured by an optical sensor. For example, the optical feature **1014** may comprise a hole. In some variations, the optical feature **1014** may comprise a piece of reflective tape. The rate of rotation of the rotor **512** may be determined by the number of times the optical feature **1014** passes by the corresponding optical sensor in a specified unit of time.

FIG. **11**A, FIG. **11**B, and FIG. **11**C provide illustrative variations of flow paths through the rotor **512**. The fluid may flow through any one of the selector valves **520** based on a selected cell process. The fluid may comprise one or more a cell solution, buffer, apheresis product, cleaning agent, or combination thereof. The selector valves may be adjusted into any one of three configuration. In a first configuration, shown in FIG. **11**A, each of the selector valves **520***a*, **520***b*, **520***c* may be in a CCW position. The flow of fluid through the rotor while the selector valves **520** are in the first configuration and the rotor **512** is rotating may perform a counterflow centrifugal elutriation process. The first configuration may be referred to as an elutriation mode. In the first configuration, fluid may flow through the fluid opening **914** in a housing **510** (not shown). Then, fluid may flow along the flow path **916** to the selector valve **520***a*, through the selector valve **520***a*, and further along a flow path **918**. Flow path **918** may fluidically connect selector valve **520***a* to the elutriation chamber **810**. In some variations, an amount of fluid and/or cells may collect within the elutriation chamber **810**. In some variations, fluid may flow from the elutriation chamber **810** and into a flow path **920**. Flow path **920** may then fork into two or more flow paths. The two or more flow paths may comprise a flow path **924** and a flow path **930**. With the selector valves in the first configuration, fluid may flow along flow path **920** and into flow path **924**. Then, fluid may flow along flow path **924** and through selector valve **520***b*. Fluid may then flow from selector valve **520***b* along flow path **928**. In some variations, flow path **928** may fluidically connect to the fluid opening **914**. In this way, fluid may flow through the flow paths of the first configuration before exiting through the fluid opening **914**. The fluid opening **914** may be fluidically connected to the liquid transfer bus. The fluid flow through the fluid opening **914** may be forward fluid flow or reverse fluid flow. In some variations, flow path **928** may fluidically connect to flow path **916**. In this way, fluid may flow through the first selector valve **520***a* such that it is recycled in a continuous path.

In a second configuration, shown in FIG. **11**B, the selector valves **520***a* and **520***b* may be in a CW position and selector valve **520***c* may be in a CCW position. The flow of fluid through the rotor **512** while the selector valves **520** are in the second configuration and the rotor **512** is rotating may perform a spinoculation process. The second configuration may be referred to as a spinoculation mode. In the second configuration, fluid may flow through the fluid opening **914** in a housing **510** (not shown). Then, fluid may flow along the flow path **916** to the selector valve **520***a*, through the selector valve **520***a*, and further along a flow path **922**. Flow path **922** may fluidically connect selector valve **520***a* to the spinoculation chamber **514**. In some variations, an amount of fluid and/or cells may collect within the spinoculation chamber **514**. In some variations, fluid may flow from the spinoculation chamber **514** and into a flow path **926**. Fluid may flow along flow path **926** and through selector valve **520***b*. Fluid may then flow from selector valve **520***b* along flow path **928**. In some variations, flow path **928** may fluidically connect to the fluid opening **914**. In this way, fluid may flow through the flow paths of the second configuration before exiting through the fluid opening **914**. The fluid opening **914** may be fluidically connected to the liquid transfer bus. The fluid flow through the fluid opening **914** may be forward fluid flow or reverse fluid flow. In some variations, flow path **928** may fluidically connect to flow path **916**. In this way, fluid may flow through the first selector valve **520***a* such that it is recycled in a continuous path.

In a third configuration, shown in FIG. **11**C, the selector valves **520***a* and **520***b* may be in a CCW position and selector valve **520***c* may be in a CW position. The flow of fluid through the rotor **512** while the selector valves are in the third configuration may perform an elutriation process and a spinoculation process in series. For example, a volume of fluid may flow through the rotor **512** to perform an elutriation process followed by a spinoculation process. The third configuration may be referred to as a lysis-free mode.

In the third configuration, fluid may flow through the fluid opening **914** in a housing **510** (not shown). Then, fluid may flow along the flow path **916** to the selector valve **520***a*, through the selector valve **520***a*, and further along a flow path **918**. Flow path **918** may fluidically connect selector valve **520***a* to the elutriation chamber **810**. In some variations, an amount of fluid may collect within the elutriation chamber **810**. In some variations, fluid may flow from the elutriation chamber **810** and into a flow path **920**. Flow path **920** may then fork into two or more flow paths. The two or more flow paths may comprise a flow path **924** and a flow path **930**. With the selector valves in the third configuration, fluid may flow along flow path **920** and into flow path **930**. Then, fluid may flow along flow path **930**, which is routed around the fluid opening **914**, and to selector valve **520***c*. Fluid may then flow from selector valve **520***c* along flow path **932**. Flow path **932** may fluidically connect to the spinoculation chamber **514**. In some variations, fluid may flow from the spinoculation chamber **514** and into a flow path **926**. Fluid may flow along flow path **926** and through selector valve **520***b*. Fluid may then flow from selector valve **520***b* along flow path **928**. In some variations, flow path **928** may fluidically connect to the fluid opening **914**. In this way, fluid may flow one time through the flow paths of the third configuration before exiting through the fluid opening **914**. The fluid opening **914** may be fluidically connected to the liquid transfer bus. The fluid flow through the fluid opening **914** may be forward fluid flow or reverse fluid flow. In some variations, fluid channel **928** may fluidically connect to fluid channel **916**. In this way, fluid may pass through the first selector valve **520***a* such that it is recycled in a continuous path.

In the third configuration, a buffer may be circulated through the rotor and subsequently recirculated. For example, in the lysis free mode, certain cell types or particulates within the fluid may be first captured within the elutriation chamber. Meanwhile, red blood cells may be pushed out from the elutriation chamber and subsequently captured in the spinoculation chamber. This process occurs while circulating and recirculating the buffer. The combined process advantageously reuses fluid (e.g., buffer) rather than requiring a continuous supply of fresh fluid.

II. Methods of Cell Processing

Generally, the systems and devices described herein may perform one or more cell processing steps. FIG. **12**A and FIG. **12**B are flowcharts of illustrative methods of cell processing that can be performed via the systems and devices described above. In FIG. **12**A, the method **1201** may comprise performing a cell processing step within a module of a cartridge. The method **1201** may include transferring a cartridge **114** within a workcell **110** in a step **1210**. For example, the cartridge **114** may be removed from a reagent vault **226** by a robot **116** and subsequently moved to one or more instruments within the workcell **110**. In some variations, the instrument may be a spinoculation instrument **230**. In further variations, the instrument may be a cell separation instrument **216** (e.g., magnetic separation instrument), an electroporation instrument **220**, a counterflow centrifugation elutriation (CCE) instrument **222**, or combinations thereof. In a step **1211**, cells may be transferred from a liquid transfer bus **322** to a module of the cartridge **114**. For example, a command may be sent by the controller **120** to the liquid transfer bus **322** to open at least one valve and begin flow of a fluid therefrom. In some variations, the module may comprise the module **310**. In further variations, the module may comprise a bioreactor module **318**, an electroporation module **320**, a magnetic selection module **324**, or an auxiliary module **326**. Then, in a step **1214**, a cell processing step may be performed within the module. For example, the cell processing step may comprise one or more spinoculation, elutriation, magnetic cell separation, electroporation, and any other cell process. Cells may then be transferred from the module to the liquid transfer bus **322** in a step **1216**. For example, cells may flow along at least one flow path from the module to the liquid transfer bus **322**. Cells may flow through at least one valve of the liquid transfer bus **322**. The valve status may be controlled by the controller **120**.

In FIG. **12**B, the method **1202** may comprise performing a cell processing step within a sub-module of a module of a cartridge. The method **1202** may include transferring a cartridge **114** within a workcell **110** in a step **1210**. For example, the cartridge **114** may be removed from a reagent vault **226** by a robot **116** and subsequently moved to one or more instruments within the workcell **110**. In some variations, the instrument may be a spinoculation instrument **230**. In further variations, the instrument may be a cell separation instrument **216** (e.g., magnetic separation instrument), an electroporation instrument **220**, or a counterflow centrifugation elutriation (CCE) instrument **222**, or combinations thereof. In a step **1212**, cells may be transferred from a liquid transfer bus **322** to a sub-module of the cartridge **114**. For example, a command may be sent by the controller **120** to the liquid transfer bus **322** to open at least one valve and begin flow of a fluid therefrom. In some variations, the sub-module may comprise a sub-module **314** of the module **310**. The sub-module **314** may comprise a spinoculation sub-module. In some variations, the sub-module may comprise a sub-module **316** of the module **310**. The sub-module **314** may comprise an elutriation sub-module. Then, in a step **1218**, a cell processing step may be performed within the sub-module. For example, the cell processing step may comprise spinoculation within the spinoculation sub-module **314**. Alternatively, in a step **1220**, a cell processing step may be performed within the sub-module. For example, the cell processing step may comprise elutriation within the elutriation sub-module **316**. Cells may then be transferred from the module to the liquid transfer bus **322** in a step **1216**. Cells may flow through at least one valve of the liquid transfer bus **322**. The valve status may be controlled by the controller **120**.

FIG. **13**A and FIG. **13**B are flowcharts of illustrative methods of cell processing that can be performed via the systems and devices described above. In FIG. **13**A, the method **1301** may comprise circulating a buffer while performing a cell processing step. The method **1301** may include transferring a cartridge within a workcell in a step **1210**. For example, the cartridge **114** may be removed from a reagent vault **226** by a robot **116** and subsequently moved to one or more instruments within the workcell **110**. In some variations, the instrument may be a spinoculation instrument **230**. In further variations, the instrument may be a cell separation instrument **216** (e.g., magnetic separation instrument), an electroporation instrument **220**, a counterflow centrifugation elutriation (CCE) instrument **222**, or combinations thereof. In a step **1212**, cells may be transferred from a liquid transfer bus **322** to a sub-module of the cartridge **310**. For example, a command may be sent by the controller **120** to the liquid transfer bus **322** to open at least one valve and begin flow of a fluid therefrom. In some variations, the sub-module may comprise a sub-module **314** of the module **310**. The sub-module **314** may comprise a spinoculation sub-module. In some variations, the sub-module may comprise a sub-module **316** of the module **310**. The sub-module **314** may comprise an elutriation sub-module. Then, a buffer may begin to circulate through at least one sub-module in a step **1310**. For example, the buffer may flow from the liquid transfer bus **322**, flow through a fluid opening **526** of the module **310**, and through at least one of the sub-modules **314**, **316**. While the buffer is circulating, a spinoculation process may be performed in a step **1312**. For example, the spinoculation process may be performed within the spinoculation sub-module **314**. Alternatively, while the buffer is circulating, an elutriation process may be performed in a step **1314**. For example, the elutriation process may be performed within the elutriation sub-module **316**. Subsequently, cells may be transferred from the module to the liquid transfer bus **322** in a step **1216**. Cells may flow through at least one valve of the liquid transfer bus **322**. The valve status may be controlled by the controller **120**. In some variations, the buffer may also flow through at least one valve of the liquid transfer bus **322**.

In FIG. **13**B, the method **1302** may comprise performing cell processing steps while recirculating a buffer. The method **1302** may include transferring a cartridge within a workcell in a step **1210**. For example, the cartridge **114** may be removed from a reagent vault **226** by a robot **116** and subsequently moved to one or more instruments within the workcell **110**. In some variations, the instrument may be a spinoculation instrument **230**. In further variations, the instrument may be a cell separation instrument **216** (e.g., magnetic separation instrument), an electroporation instrument **220**, a counterflow centrifugation elutriation (CCE) instrument **222**, or combinations thereof. In a step **1212**, cells may be transferred from a liquid transfer bus **322** to a sub-module of the cartridge **310**. For example, a command may be sent by the controller **120** to the liquid transfer bus **322** to open at least one valve and begin flow of a fluid therefrom. In some variations, the sub-module may comprise a sub-module **314** of the module **310**. The sub-module **314** may comprise a spinoculation sub-module. In some variations, the sub-module may comprise a sub-module **316** of the module **310**. The sub-module **314** may comprise an elutriation sub-module. Then, a buffer may begin to circulate through at least one sub-module in a step **1310**. For example, the buffer may flow from the liquid transfer bus **322**, flow through a fluid opening **526** of the module **310**, and through at least one of the sub-modules **314**, **316**. In a step **1318**, the buffer may be recirculated through at least one of sub-module. For example, at least one valve of the liquid transfer bus **322** may be closed in response to a command sent by the controller **120**. The closed valve of the liquid transfer bus **322** may prevent buffer from exiting the module **310**. There may be at least one fluid path along which the buffer may recirculate through the at least one sub-module. While the buffer is recirculating, a spinoculation process may be performed in a step **1312**. For example, the spinoculation process may be performed within the spinoculation sub-module **314**. Additionally, while the buffer is recirculating, an elutriation process may be performed in a step **1314**. For example, the elutriation process may be performed within the elutriation sub-module **316**. Subsequently, cells may be transferred from the module to the liquid transfer bus **322** in a step **1216**. Cells may flow through at least one valve of the liquid transfer bus **322**. The valve status may be controlled by the controller **120**. In some variations, the buffer may also flow through at least one valve of the liquid transfer bus **322**.

FIG. **14**A, FIG. **14**B, and FIG. **14**C are flowcharts of illustrative methods of cell processing that can be performed via the systems and devices described above. In FIG. **14**A, a method **1401** may comprise performing a cell processing step after adjusting at least one selector valve. The method **1401** may include a cell processing step may be selected in a step **1410**. For example, a user may select at least one cell processing step via the controller **120** of the workcell **110**. In some variations, the user may select multiple cell processing steps. For example, the user may select one or more of elutriation and spinoculation. In some variations, the cartridge **114** may already be coupled to at least one instrument. For example, the cartridge may already be coupled to one or more of a spinoculation instrument and elutriation instrument. Based on the selected cell processing step, at least one selector valve may be adjusted in a step **1412**. For example, the position of at least one selector valve **520** of a module **310** may correspond to at least one cell processing step. For example, the first configuration of the selector valves **520** may correspond to an elutriation process and the second configuration of the selector valves **520** may correspond to a spinoculation process. In some variations, the position of at least one selector valve **520** may not require adjustment if the selector valve **520** is already in the configuration required to perform the selected cell processing step. In further variations, adjustment of at least one selector valve **520** may be required to perform the selected cell processing step. For example, the gear **530** of the workcell **110** may rotate in response to the selected cell process. In turn, the rotation of gear **530** rotates the gear **528** of the module **310**, which subsequently may adjust one or more selector valve **520**.

Then, cells may be transferred from a liquid transfer bus through the at least one selector valve to a sub-module in a step **1414**. For example, a command may be sent by the controller **120** to the liquid transfer bus **322** to open at least one valve and begin flow of a fluid therefrom. In some variations, the sub-module may comprise a sub-module **314** of the module **310**. The sub-module **314** may comprise a spinoculation sub-module. In some variations, the sub-module may comprise a sub-module **316** of the module **310**. The sub-module **314** may comprise an elutriation sub-module. A cell processing step may be performed in a step **1416**. For example, the spinoculation process may be performed within the spinoculation sub-module **314**. In another example, the elutriation process may be performed within the elutriation sub-module **316**. Subsequently, cells may be transferred from the sub-module through at least one selector valve to the liquid transfer bus in a step **1418**. Cells may flow through at least one valve of the liquid transfer bus **322**. The valve status may be controlled by the controller **120**.

In FIG. **14**B, a method **1402** may comprise performing multiple cell process steps in series after adjusting at least one selector valve. The method **1402** may include a cell processing step may be selected in a step **1410**. For example, a user may select at least one cell processing step via the controller **120** of the workcell **110**. In some variations, the user may select multiple cell processing steps. For example, the user may select one or more of elutriation and spinoculation. In some variations, the cartridge **114** may already be coupled to at least one instrument. For example, the cartridge may already be coupled to one or more of a spinoculation instrument and elutriation instrument. Based on the selected cell processing step, at least one selector valve may be adjusted in a step **1412**. For example, the position of at least one selector valve **520** of a module **310** may correspond to at least one cell processing step. For example, the first configuration of the selector valves **520** may correspond to an elutriation process, the second configuration of the selector valves **520** may correspond to a spinoculation process, and the third configuration of the selector valves **520** may correspond to an elutriation process and spinoculation process in series. In some variations, the position of at least one selector valve **520** may not require adjustment if the selector valve **520** is already in the configuration required to perform the selected cell processing step. In further variations, adjustment of at least one selector valve **520** may be required to perform the selected cell processing step. For example, the gear **530** of the workcell **110** may rotate in response to the selected cell process. In turn, the rotation of gear **530** rotates the gear **528** of the module **310**, which subsequently may adjust one or more selector valve **520**.

Then, cells may be transferred from a liquid transfer bus through the at least one selector valve to one or more sub-module in a step **1420**. For example, a command may be sent by the controller **120** to the liquid transfer bus **322** to open at least one valve and begin flow of a fluid therefrom. In some variations, the sub-module may comprise a sub-module **314** of the module **310**. The sub-module **314** may comprise a spinoculation sub-module. In some variations, the sub-module may comprise a sub-module **316** of the module **310**. The sub-module **314** may comprise an elutriation sub-module. In some variations, the cells may be transferred from the liquid transfer bus **322** to both sub-modules **314**, **316**. Then, elutriation may be performed in a sub-module **316** in a step **1422**. In some variations, spinoculation may be subsequently performed in a sub-module **314** in a step **1424**. For example, the elutriation and spinoculation process steps may occur in series. Subsequently, cells may be transferred from the sub-modules **314**, **316** through at least one selector valve to the liquid transfer bus in a step **1418**. Cells may flow through at least one valve of the liquid transfer bus **322**. The valve status may be controlled by the controller **120**.

In FIG. **14**C, a method **1403** may comprise performing multiple cell process steps in series after adjusting at least one selector valve and measuring at least one parameter of the module **310**. The method **1402** may include a cell processing step may be selected in a step **1410**. For example, a user may select at least one cell processing step via the controller **120** of the workcell **110**. In some variations, the user may select multiple cell processing steps. For example, the user may select one or more of elutriation and spinoculation. In some variations, the cartridge **114** may already be coupled to at least one instrument. For example, the cartridge may already be coupled to one or more of a spinoculation instrument and elutriation instrument. Based on the selected cell processing step, at least one selector valve may be adjusted in a step **1412**. For example, the position of at least one selector valve **520** of a module **310** may correspond to at least one cell processing step. For example, the first configuration of the selector valves **520** may correspond to an elutriation process, the second configuration of the selector valves **520** may correspond to a spinoculation process, and the third configuration of the selector valves **520** may correspond to an elutriation process and spinoculation process in series. In some variations, the position of at least one selector valve **520** may not require adjustment if the selector valve **520** is already in the configuration required to perform the selected cell processing step. In further variations, adjustment of at least one selector valve **520** may be required to perform the selected cell processing step. For example, the gear **530** of the workcell **110** may rotate in response to the selected cell process. In turn, the rotation of gear **530** rotates the gear **528** of the module **310**, which subsequently may adjust one or more selector valve **520**.

Then, cells may be transferred from a liquid transfer bus through the at least one selector valve to one or more sub-module in a step **1420**. For example, a command may be sent by the controller **120** to the liquid transfer bus **322** to open at least one valve and begin flow of a fluid therefrom. In some variations, the sub-module may comprise a sub-module **314** of the module **310**. The sub-module **314** may comprise a spinoculation sub-module. In some variations, the sub-module may comprise a sub-module **316** of the module **310**. The sub-module **314** may comprise an elutriation sub-module. In some variations, the cells may be transferred from the liquid transfer bus **322** to both sub-modules **314**, **316**. Then, elutriation may be performed in a sub-module **316** in a step **1422**. In some variations, spinoculation may be subsequently performed in a sub-module **314** in a step **1424**. For example, the elutriation and spinoculation process steps may occur in series when the selector valves **520** are in the third configuration. The rate of rotation of the at least one sub-module may be measured in a step **1428**. For example, the sensor **140*b*** may be operatively coupled to the window **532*b*** to measure an optical feature of the protrusion **910** as the protrusion **910** rotates past the sensor **140*b*** to measure the rate of rotation. Imaging data of the elutriation chamber **516** of the elutriation sub-module **316** may be generated in a step **1430**. For example, the sensor **140*a*** may be operatively coupled to the window **532*a*** to measure one or more of the quantity and position of cells within the elutriation chamber **516**. Subsequently, cells may be transferred from the sub-modules **314**, **316** through at least one selector valve to the liquid transfer bus in a step **1418**. Cells may flow through at least one valve of the liquid transfer bus **322**. The valve status may be controlled by the controller **120**.

While described above as containing certain steps, it should be understood that the methods of cell processing may include any subset of cell processing steps in any suitable order.

All references cited are herein incorporated by reference in their entirety.

Throughout this application, the term "about" is used to indicate that a value includes the inherent variation of error for the device or the method being employed to determine the value, or the variation that exists among the samples being measured. Unless otherwise stated or otherwise evident from the context, the term "about" means within 10% above or below the reported numerical value (except where such number would exceed 100% of a possible value or go below 0%). When used in conjunction with a range or series of values, the term "about" applies to the endpoints of the range or each of the values enumerated in the series, unless otherwise indicated. As used in this application, the terms "about" and "approximately" are used as equivalents.

While embodiments of the present invention have been shown and described herein, those skilled in the art will understand that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the invention. It should be understood that various alternatives to the embodiments of the invention described herein may be employed in practicing the invention. It is intended that the following claims define the scope of the invention and that methods and structures within the scope of these claims and their equivalents be covered thereby.

What is claimed is:

1. A cartridge for processing cells in an automated system comprising:
- a liquid transfer bus;
- a module fluidically coupled to the liquid transfer bus, wherein the module comprises:
  - a housing comprising a first portion and a second portion configured to couple together to form a cavity;
  - a spinoculation sub-module; and
  - an elutriation sub-module, wherein the spinoculation and elutriation sub-modules are enclosed within the cavity formed by the first and second portions of the housing; and
  - a rotor comprising at least one integrated selector valve configured to direct fluid to at least one of the spinoculation and elutriation sub-modules.

2. The cartridge of claim **1**, wherein the at least one selector valve comprises a first selector valve and a second selector valve.

3. The cartridge of claim **2**, further comprising a third selector valve.

4. The cartridge of claim **3**, wherein the first selector valve is adjustable to define a first fluid flow path, the second selector valve is adjustable to define a second fluid flow path, and the third selector valve is adjustable to define a third fluid flow path.

5. The cartridge of claim **4**, wherein the first and second selector valves are adjustable to direct fluid to a spinoculation chamber of the spinoculation sub-module.

6. The cartridge of claim **4**, wherein the first and second selector valves are adjustable to direct fluid to an elutriation chamber of the elutriation sub-module.

7. The cartridge of claim **4**, wherein the first, second, and third selector valves are adjustable to direct fluid to an elutriation chamber of the elutriation sub-module and fluid to a spinoculation chamber of the spinoculation sub-module in series.

8. The cartridge of claim **1** further comprising a first fluid opening fluidically coupled to the at least one selector valve via at least one fluid channel.

9. The cartridge of claim **8**, wherein the first fluid opening is fluidically coupled to the liquid transfer bus.

10. The cartridge of claim **1**, wherein the spinoculation sub-module and the elutriation sub-module share the rotor.

11. The cartridge of claim **1**, wherein the spinoculation sub-module comprises a spinoculation chamber having a first wall and a second wall that define a gap therebetween.

12. The cartridge of claim **1**, wherein the elutriation sub-module comprises an elutriation chamber having an elutriation cone.

13. The cartridge of claim **1**, wherein the at least one selector valve comprises a detent configured to prevent rotation of the at least one selector valve.

14. The cartridge of claim **1**, wherein the rotor further comprises an optical feature configured to be detected by an optical sensor external to the cartridge.

15. A cartridge for processing cells in an automated system comprising:

a liquid transfer bus; and

a module comprising:

a spinoculation sub-module;

an elutriation sub-module; and

a rotor positioned between and shared by the spinoculation and elutriation sub-modules, wherein the rotor comprises at least one integrated selector valve configured to direct fluid to at least one of the spinoculation and the elutriation sub-modules.

16. The cartridge of claim **15**, wherein the module further comprises a housing, and wherein the rotor, the spinoculation sub-module, and the elutriation sub-module are enclosed within a cavity formed by first and second portions of the housing.

17. The cartridge of claim **16**, wherein the housing comprises at least one viewing window.

18. The cartridge of claim **15**, wherein the at least one selector valve comprises first and second selector valves, each configured to selectively direct fluid to the spinoculation or elutriation sub-module.

19. The cartridge of claim **15**, wherein the module further comprises a seal between the spinoculation sub-module and elutriation sub-module.

20. The cartridge of claim **19**, wherein the seal comprises a first fluid opening and a second fluid opening, and wherein the first fluid opening is in fluid communication with the liquid transfer bus.

21. The cartridge of claim **20**, wherein the second fluid opening is in fluid communication with the at least one selector valve.

22. The cartridge of claim **16** further comprising a magnetic hub for attaching the cartridge to a corresponding instrument within a cell processing workcell.

23. The cartridge of claim **15**, wherein the at least one selector valve is adjustable between different positions to define different fluid flow paths through the rotor.

24. The cartridge of claim **15**, wherein the at least one selector valve comprises a rotary valve.

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